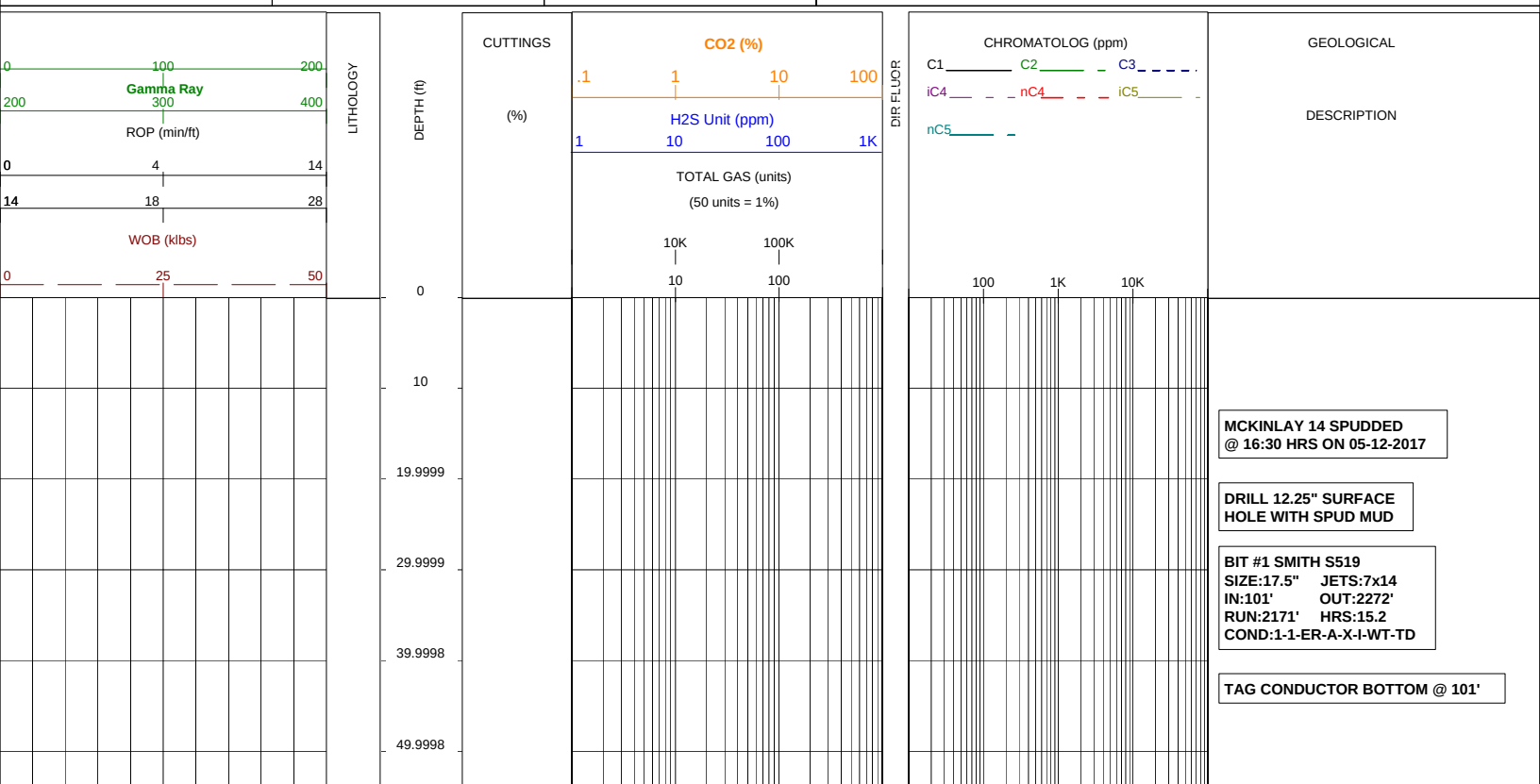
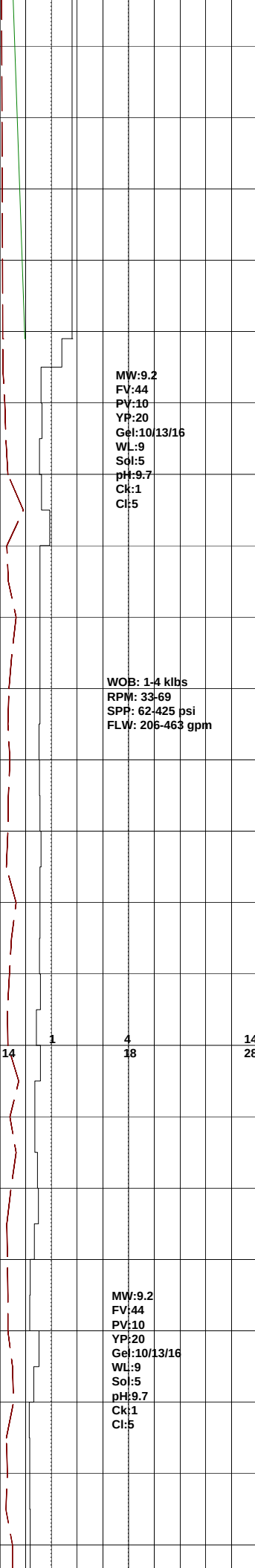


Field : MCKINLAY	Rotary Table : 141.5'	Rig : ENSIGN 971	Open Hole:	Cased Hole:	Loggers : SEAN KETTLE
Permit : PPL234	Ground Level : 118.0'	Spud Date: 05-12-2017	17.50" 2272'	13.375" 2262.5	VAHID JALEH
State : SOUTH AUSTRALIA	GDA 94 Co-ordinates :	TD Date : 17-12-2017	12.25" 5226'	9.625" 5216.9	MATHEW TOWNSEND
Country : AUSTRALIA	Lat. : 28°28'37.13"S	Total Depth : 8674'MD 4085.0'TVD	8.5" 8674'	5.5" LINER 8631.35'	
Scale : 1/240 5":100"	Long. : 140°28'58.83"E	Final Status : C & S OIL WELL			

LITHOLOGY	ACCESSORIES	DRILLING DATA	ABBREVIATIONS
 Coarse sandstone  Medium sandstone  Fine sandstone  Cement  Siltstone  Carb Siltstone  Metamorphic rock  Claystone  Limestone  Coal-lignite	 Pyrite  Siderite  Glauconite  Iron  Mica  Chert  Calcareous  Dolomitic  Carbonaceous  Fossils	 Casing shoe  Bit Trip  Wiper Trip  Core  Dst  Survey MUD DATA MW - Mud Weight (lb/gal) FV - Funnel Viscosity (s/qt) PV - Plastic Viscosity (cps) YP - Yield Point (lb/100ftsq) Gel - Gel Strength (10sec) WL - Water Loss (cc/30min) pH - Acidity / Alkalinity Ck - Cake (32nd/inch) Sol - Solids (% vol) Cl - Chlorides (mg/l)	ABBREVIATIONS BOPD - Barrels of Oil Per Day BWPD - Barrels of Water Per Day CG - Connection Gas CO - Circulate Out COND - Condensate c/c - Crush Cut DST - Drill Stem Test FLOW - Flow Rate (gal/min) GCM - Gas Cut Mud GCW - Gas Cut Water GTS - Gas To Surface INJ - Injection of Mist (bbls/hr) LCM - Lost Circulation Material MMCFD - Million Cubic Feet / Day NGTS - No Gas To Surface NOTS - No Oil To Surface NR - No Returns OCM - Oil Cut Mud OG - Over Gauge OH - Open Hole OTS - Oil To Surface Q - Flow Rate REC - Recovery Rmf - Resistivity mud filtrate ROP - Rate Of Penetration RPM - Revolutions Per Minute RTSTM - Rate Too Small To Measure Rw - Resistivity water r/r - Ring Residue SCFM - Standard Cubic Ft/Min (air) SGCM - Slightly Gas Cut Mud SPM - Strokes Per Minute SPP - Stand Pipe Pressure SWC - Side-Wall Core TG - Trip Gas WOB - Weight On Bit

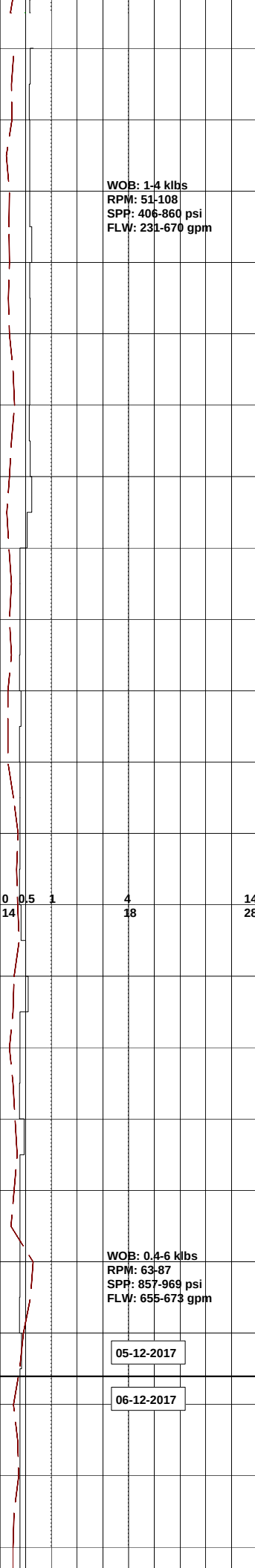




59.9998
69.9997
79.9997
89.9996
100
(101)
110
120
130
140
150
160
170
180
190
200
(205)
210
220
230
240
250
260
270

NO GAS

NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE



280
290
300
(305)
310
320
330
340
350
360
370
380
390
400
(405)
410
420
430
440
450
460
470
480
490

NO GAS

SURVEY @ 336.36' MDRT
INC: 0.35° AZI: 17.26°G

NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE

SURVEY @ 459.52'
INC: 0.62° AZI: 352.94°G

MW:9.2
FV:44
PV:10
YP:20
Gel:10/13/16
WL:9
Sol:5
pH:9.7
Ck:1
Cl:5

0 0.5 1 4 14
14 18 28

WOB:1-7 klbs
RPM:69-113
SPR:934-1185 psi
FLW:671-723 gpm

MW:9.2
FV:44
PV:10
YP:20
Gel:10/13/16
WL:9
Sol:5
pH:9.7
Ck:1
Cl:5

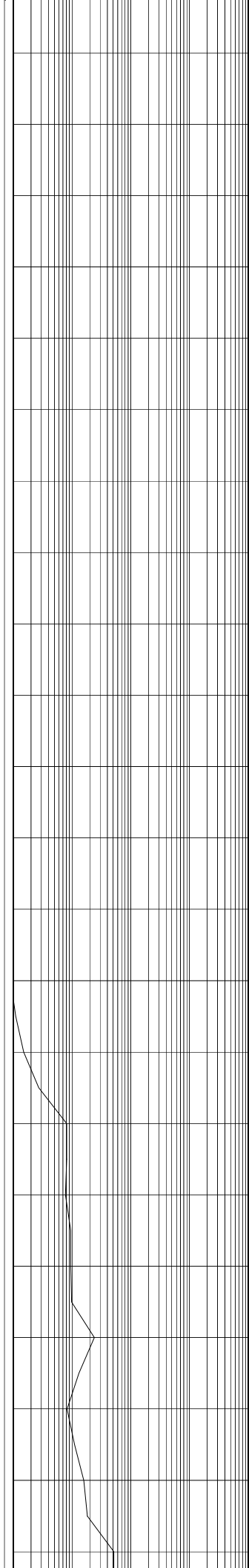
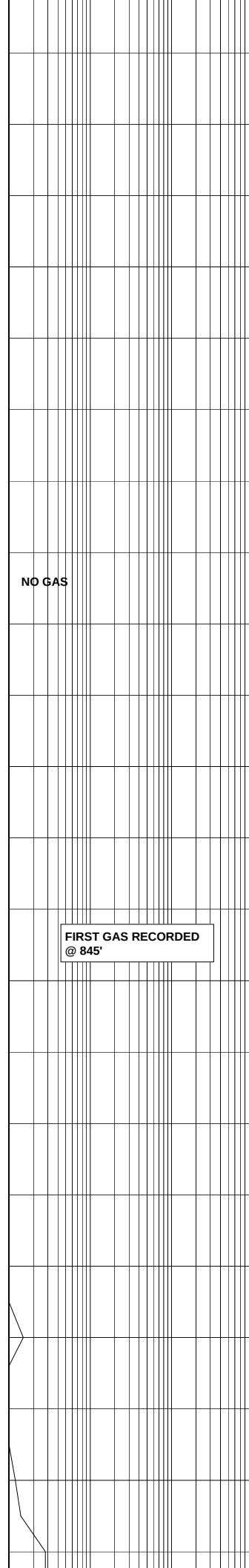
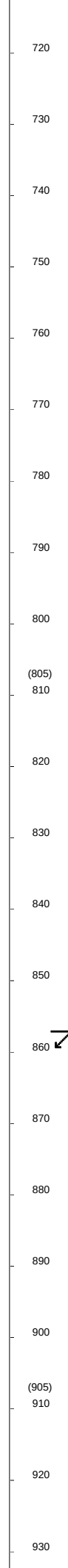
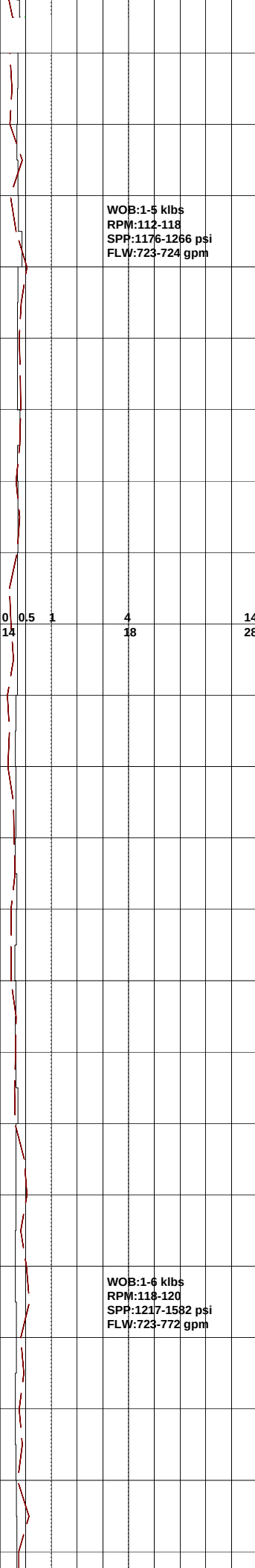
0 0.5 1 4 14
14 18 28

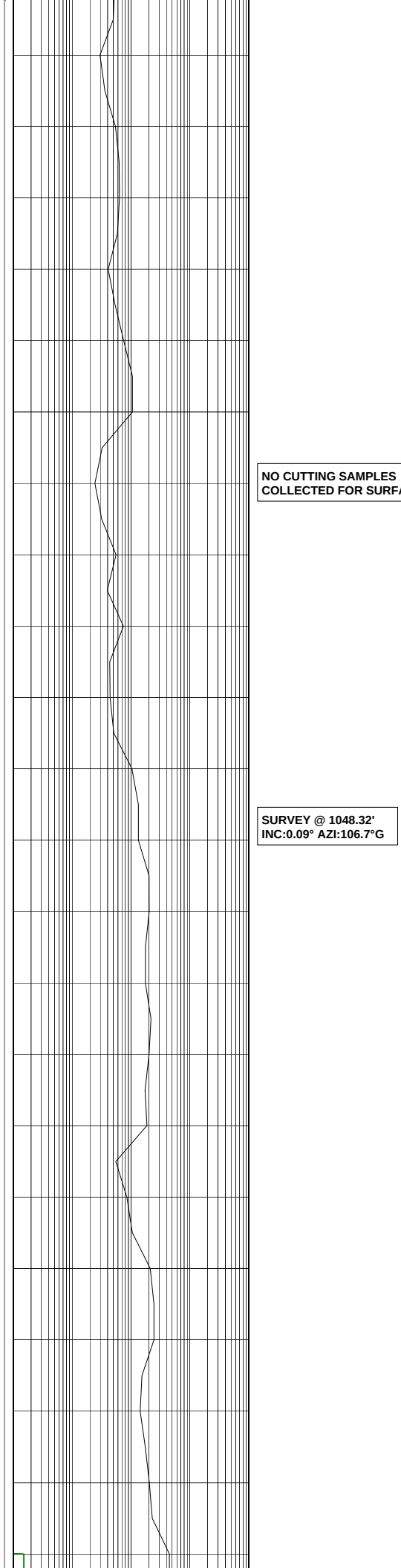
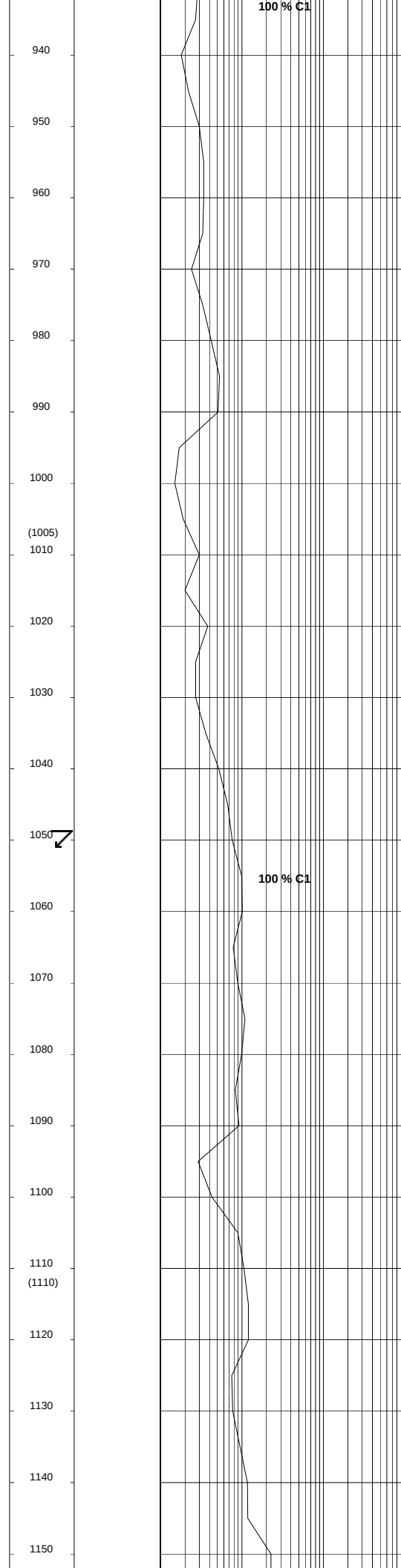
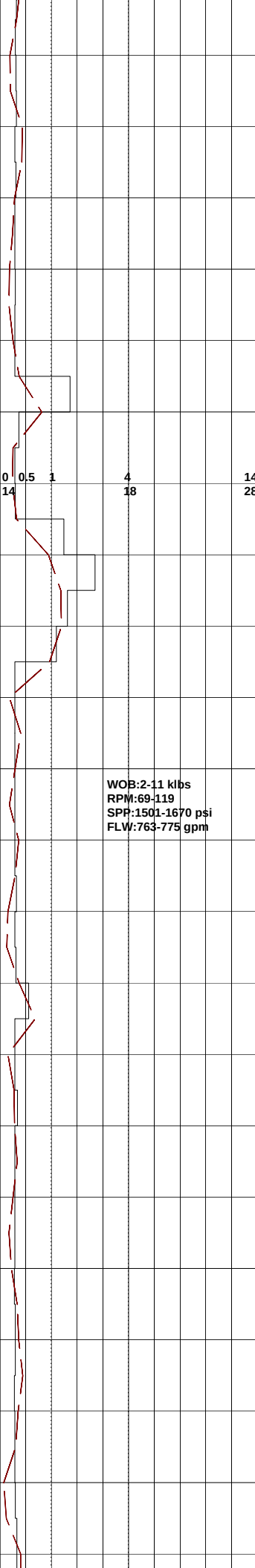
WOB:1-7 klbs
RPM:69-113
SPR:934-1185 psi
FLW:671-723 gpm

[illegible][illegible]

SURVEY @ 644.77'
INC:0.35° AZI:318.34°G

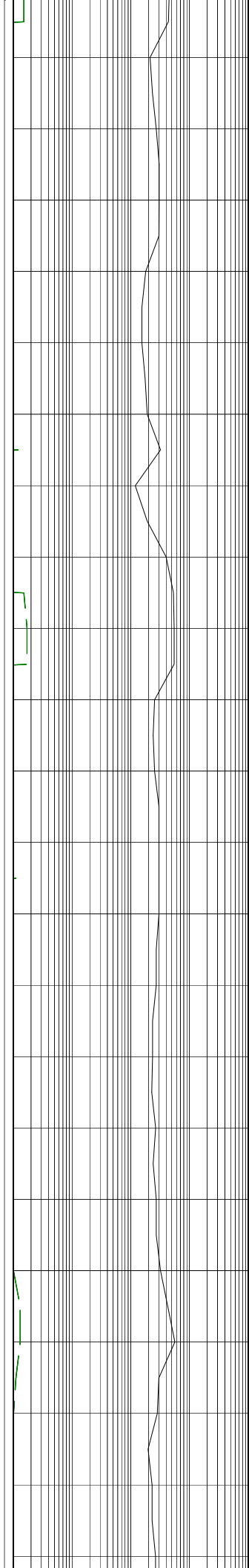
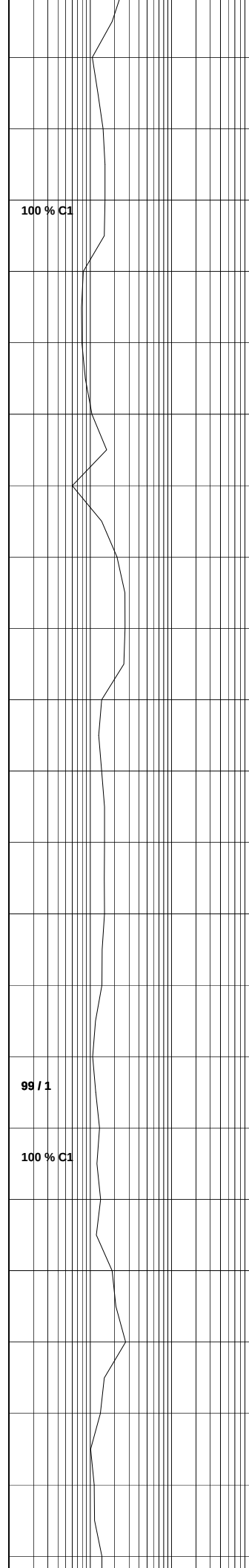
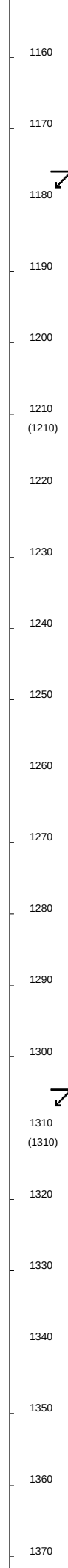
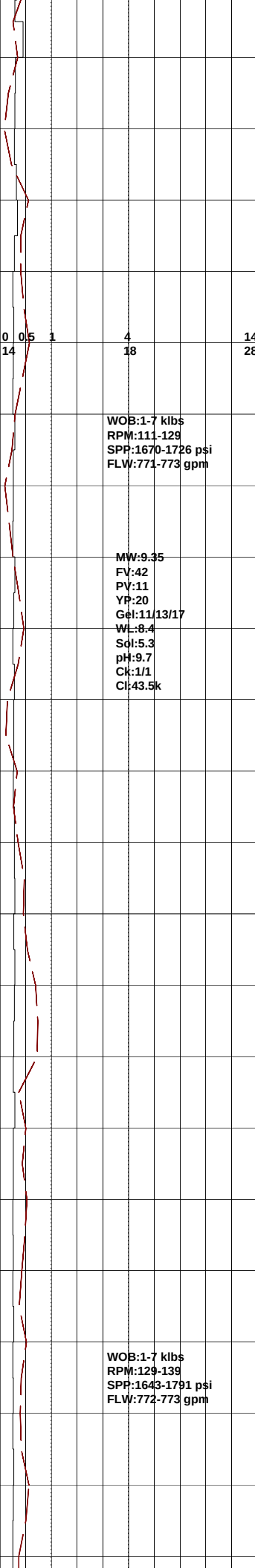
NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE





NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE

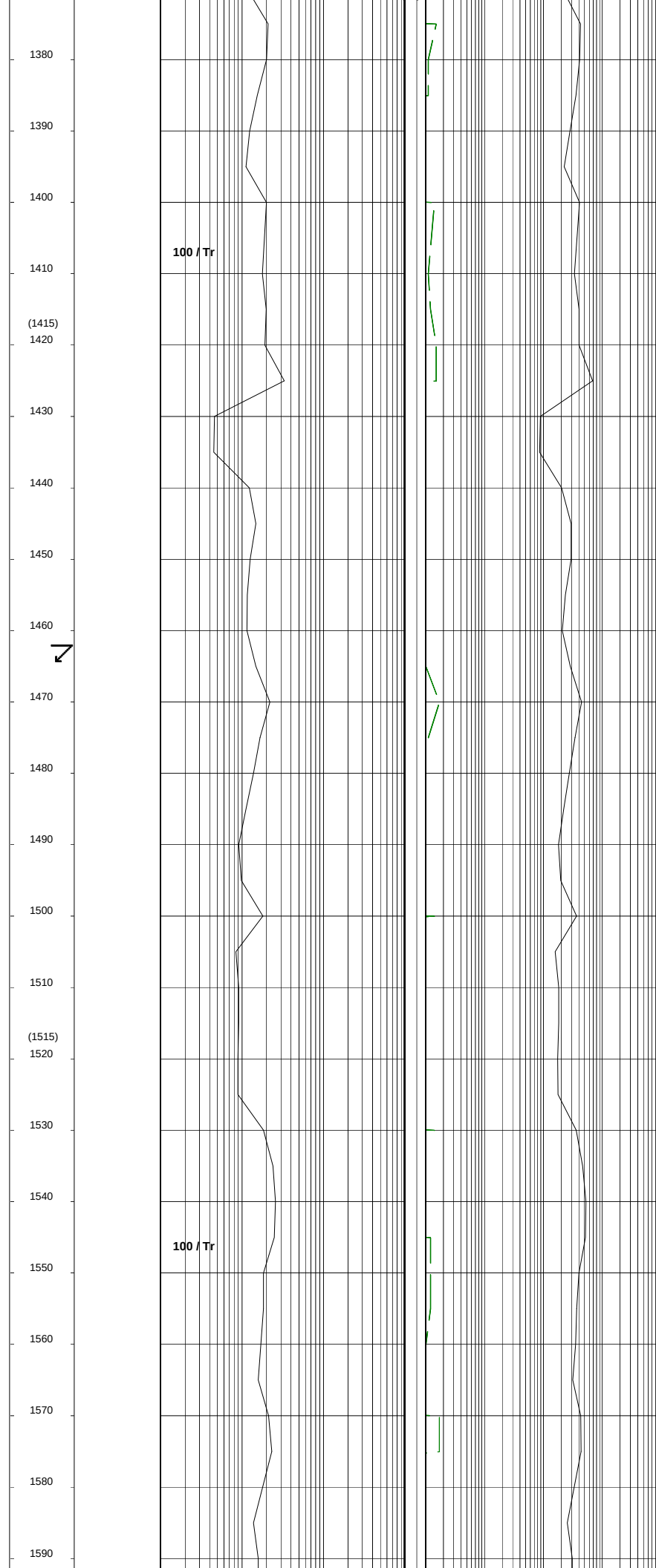
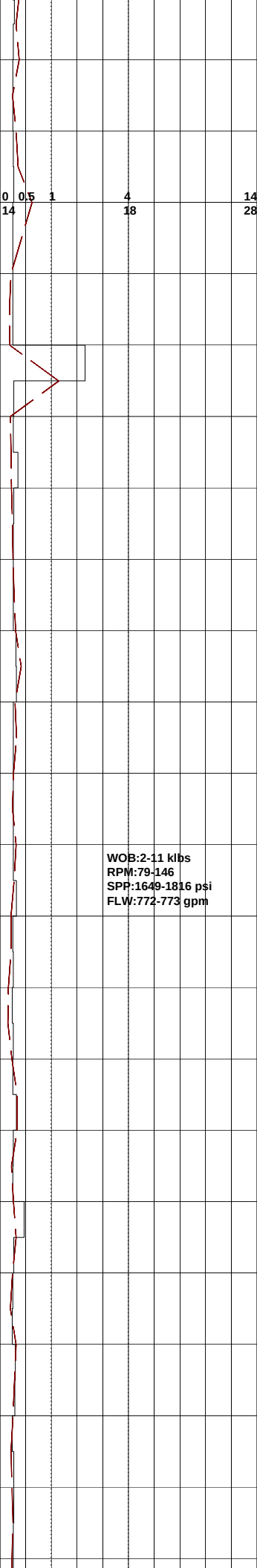
SURVEY @ 1048.32'
INC: 0.09° AZI: 106.7° G



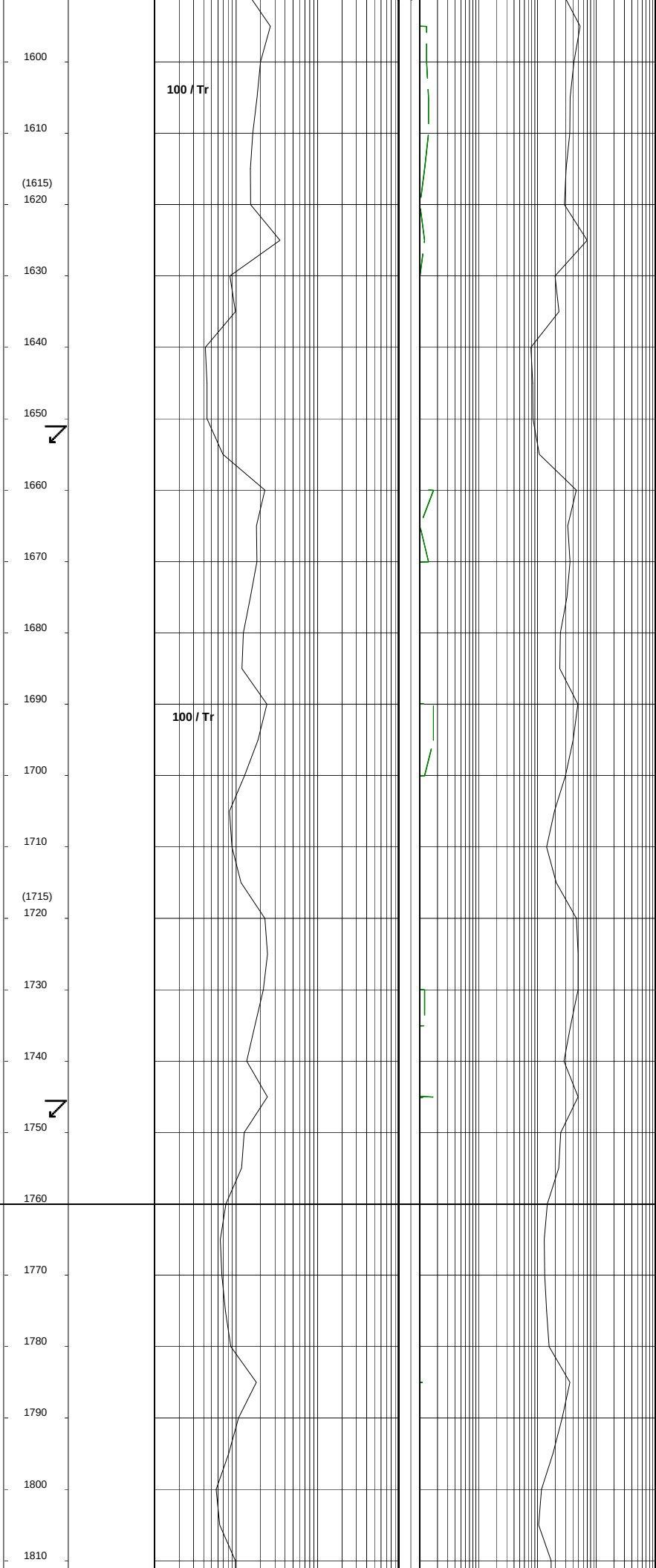
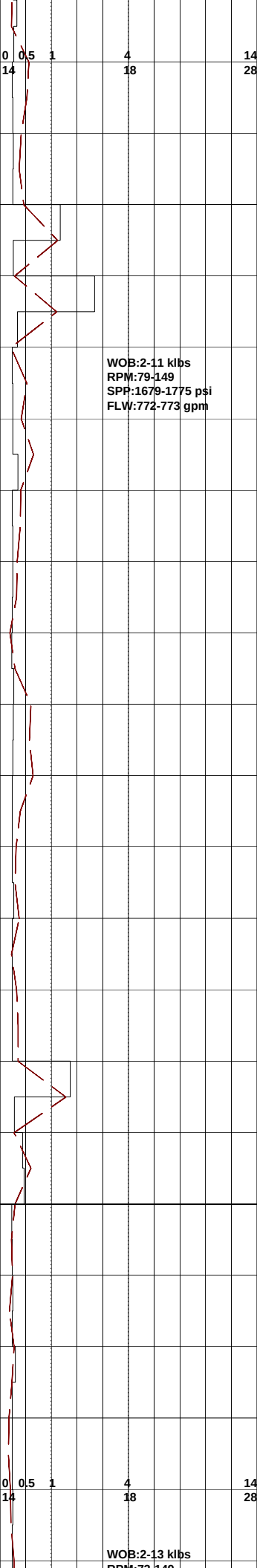
SURVEY @ 1175.92'
INC:0.35° AZI:127.5°G

NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE

SURVEY @ 1303.89'
INC:0.35° AZI:102.08°G



SURVEY @ 1461.56'
INC: 0.09° AZI: 88.62°G



NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE

SURVEY @ 1650.90' MDRT
INC: 0.09° AZI: 165.85°G

SURVEY @ 1746.02' MDRT
INC: 0.44° AZI: 131.27°G

MACKUNDA FORMATION:
1760' MDRT (-1619' TVDSS)

RPM:73-149
SPP:1732-1795 psi
FLW:772 gpm

(1815)
1820

100 / Tr

1830

1840

1850

100 / Tr

1860

1870

1880



1890

1900

SURVEY @ 1874.89'MDRT
INC: 0.35° AZI: 124.81°G

NO CUTTING SAMPLES
COLLECTED FOR SURFACE HOLE

1910

MW:9.4
FV:44
PV:13
YP:20
Gel:11/13/18
WL:8.1
Sol:5.5
pH:9.6
Ck:1/1
Cl:43k

(1920)
1920

1930

1940

WOB:1-12 klbs
RPM:63-149
SPP:1678-1761 psi
FLW:771 gpm

1950

1960

1970

1980

1990

0 0.5 1 4 14
14 18 28

2000



SURVEY @ 2000.47'MDRT
INC: 0.26° AZI: 154.02°G

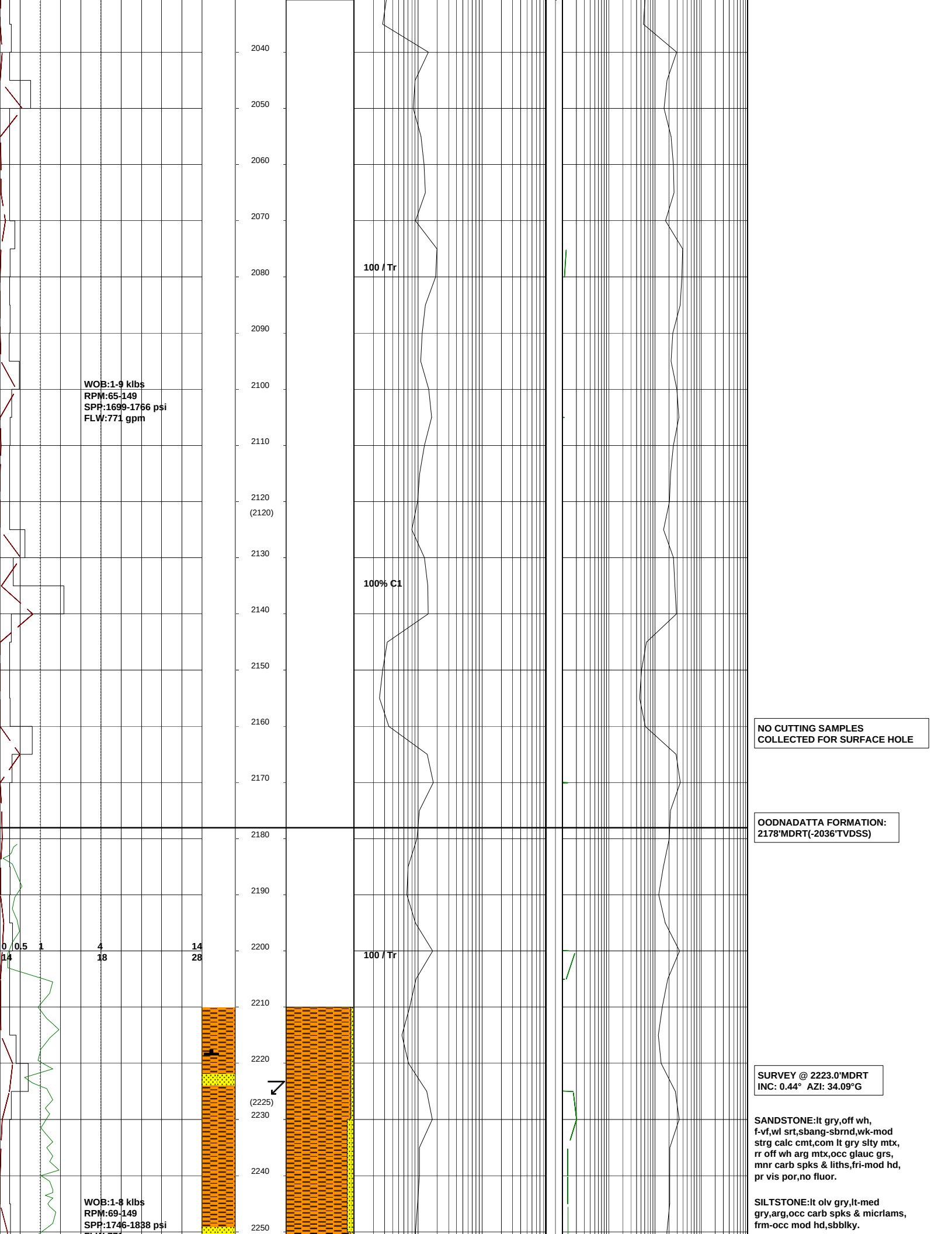
2010

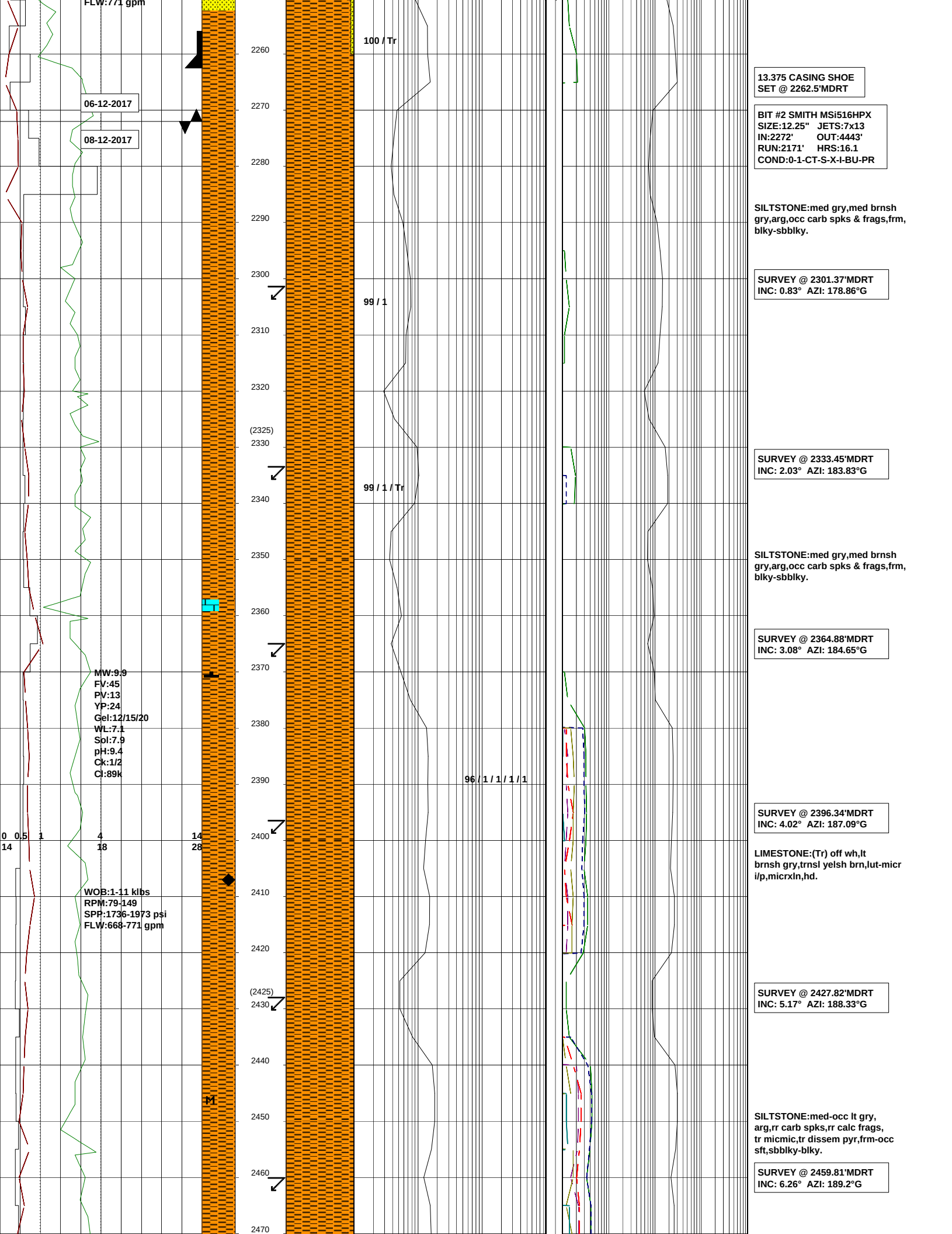
100 / Tr

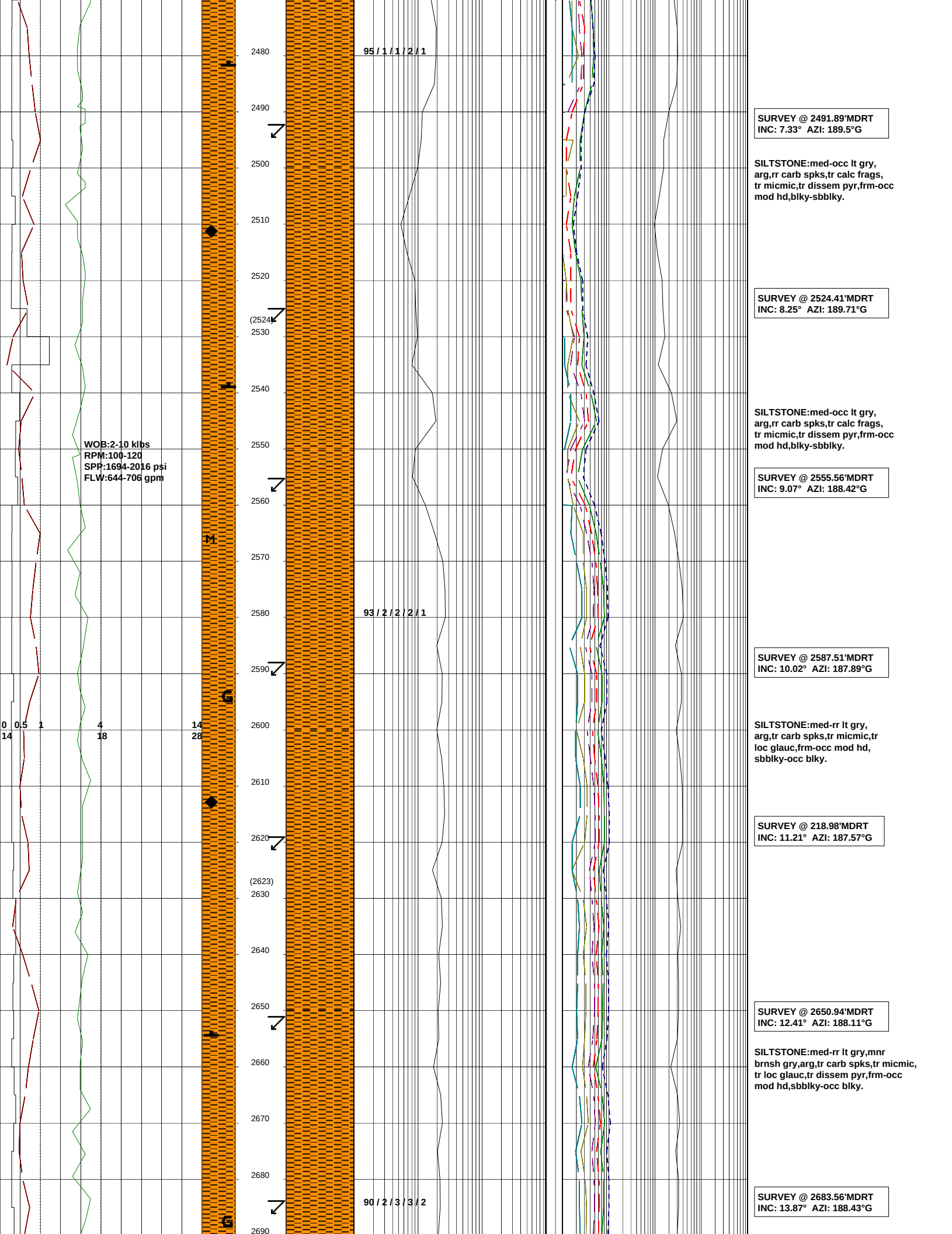
2020

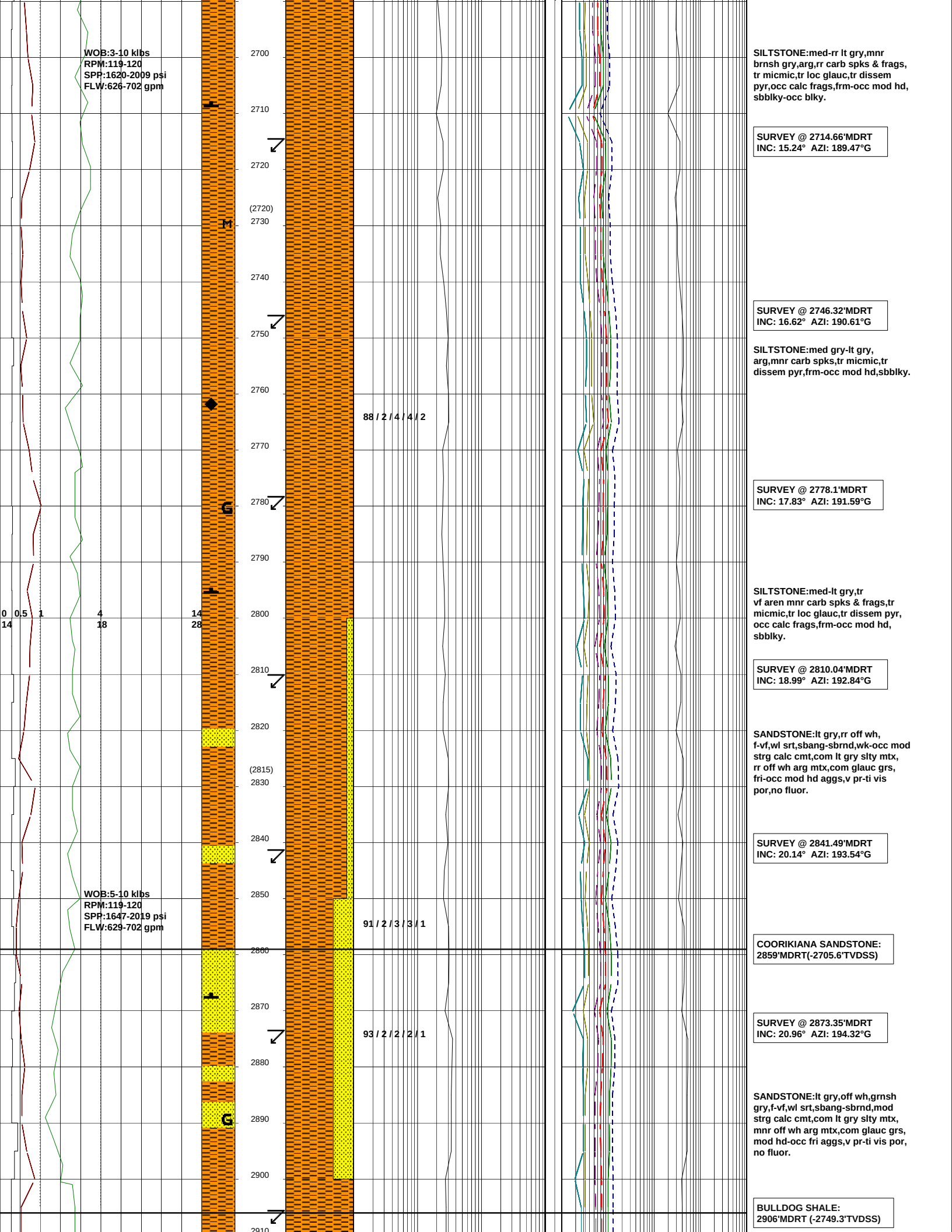
(2020)

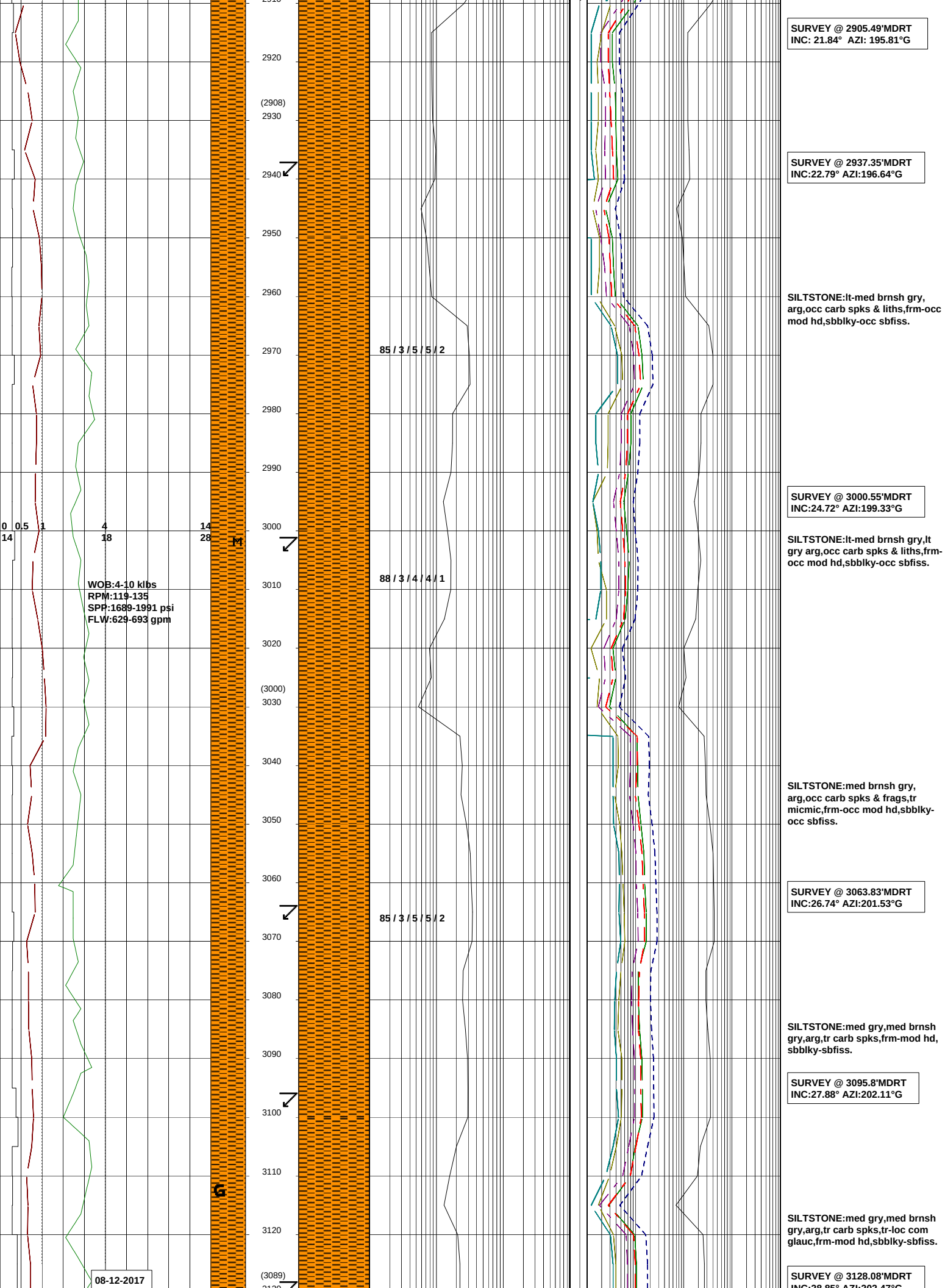
2030

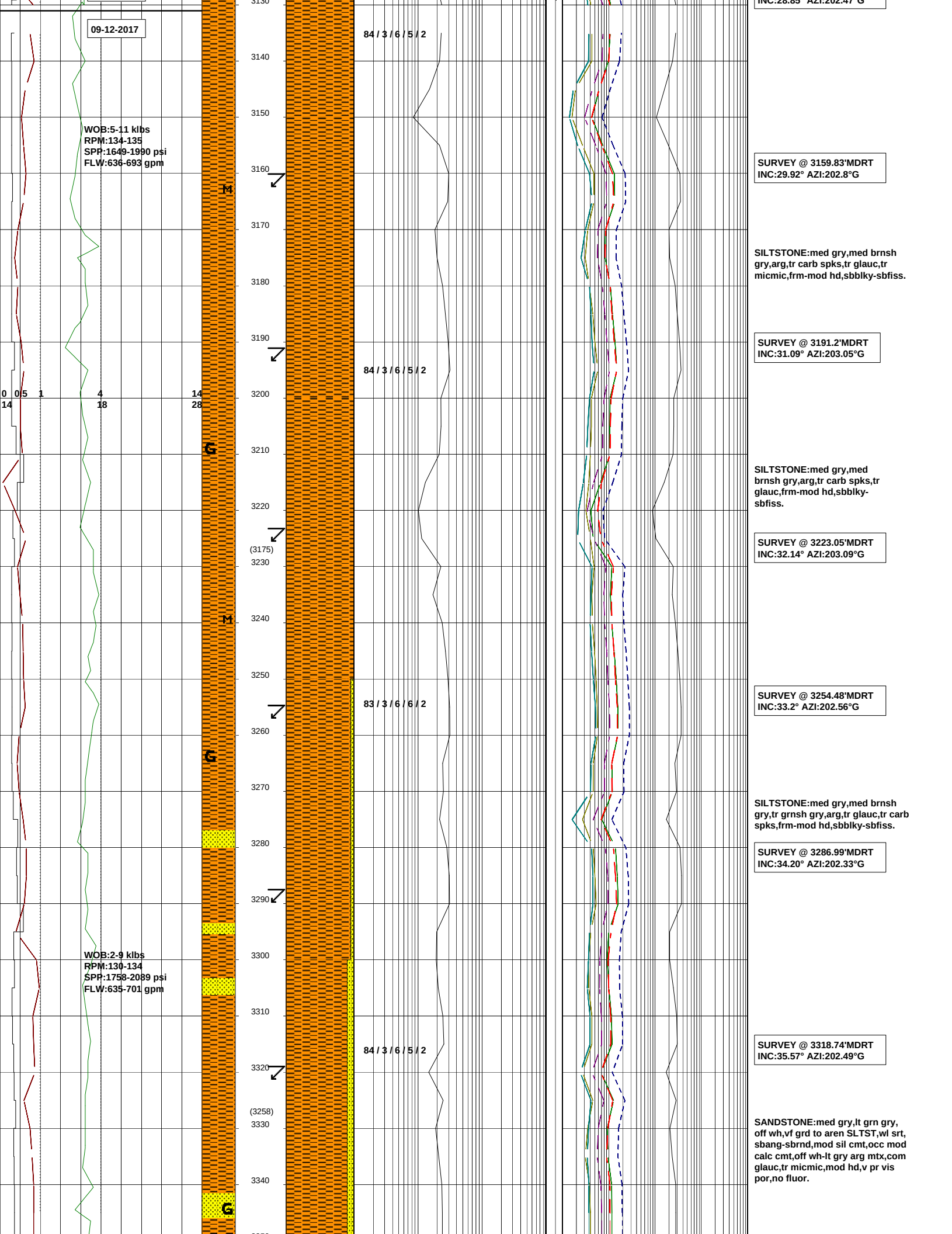


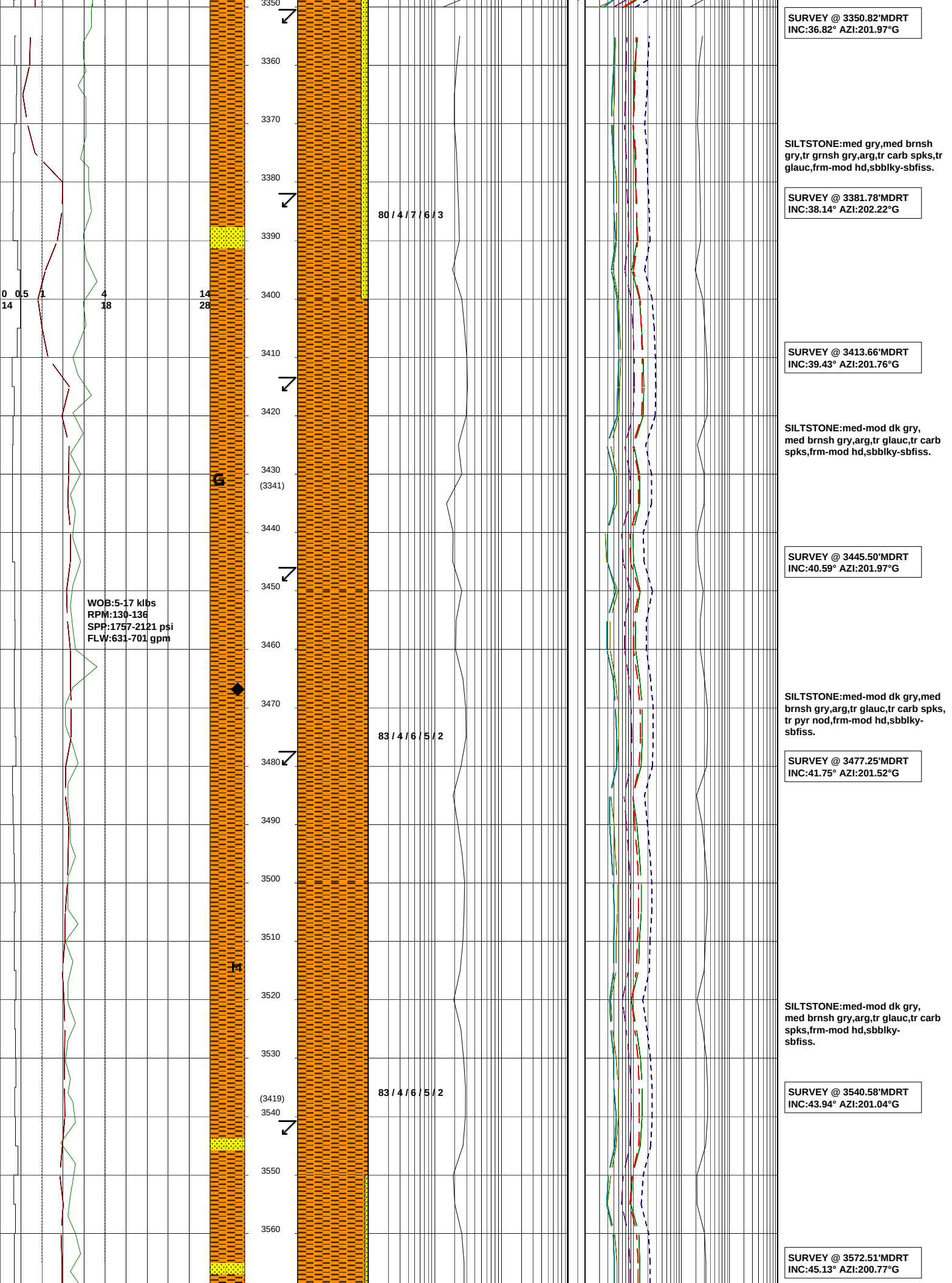


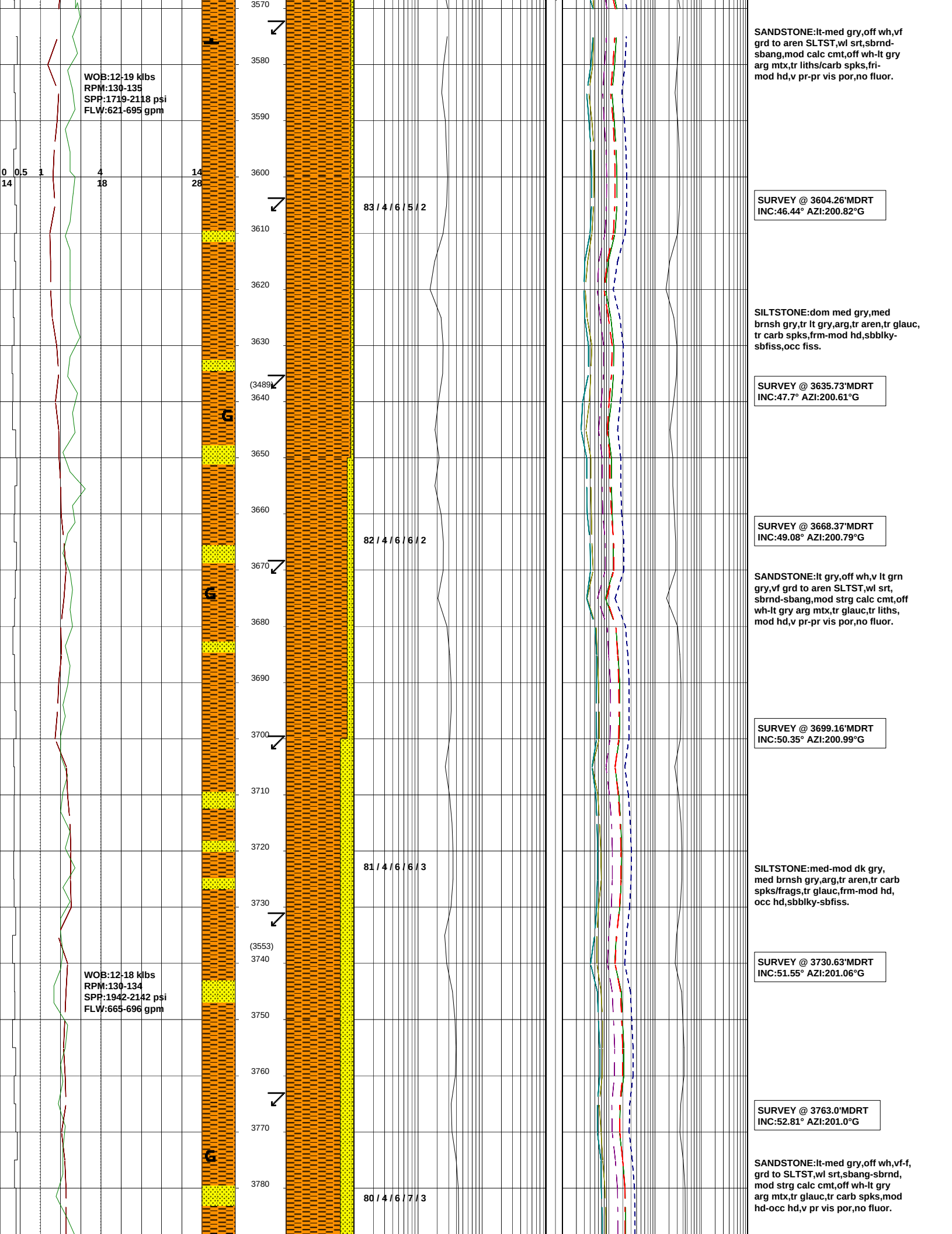


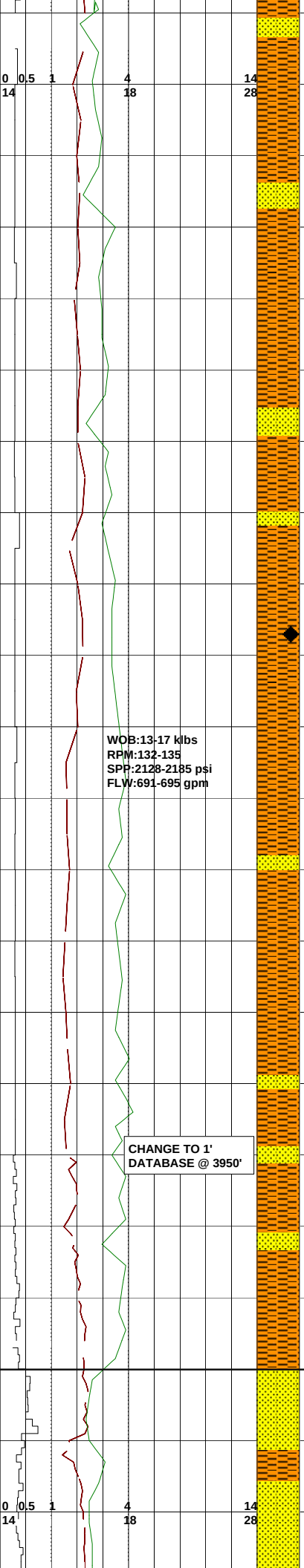






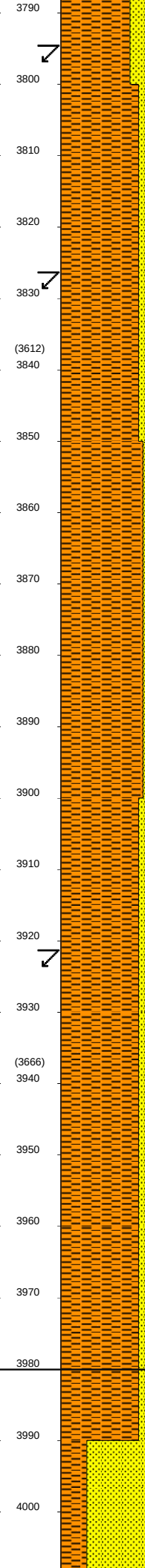






WOB:13-17 klbs
RPM:132-135
SPP:2128-2185 psi
FLW:691-695 gpm

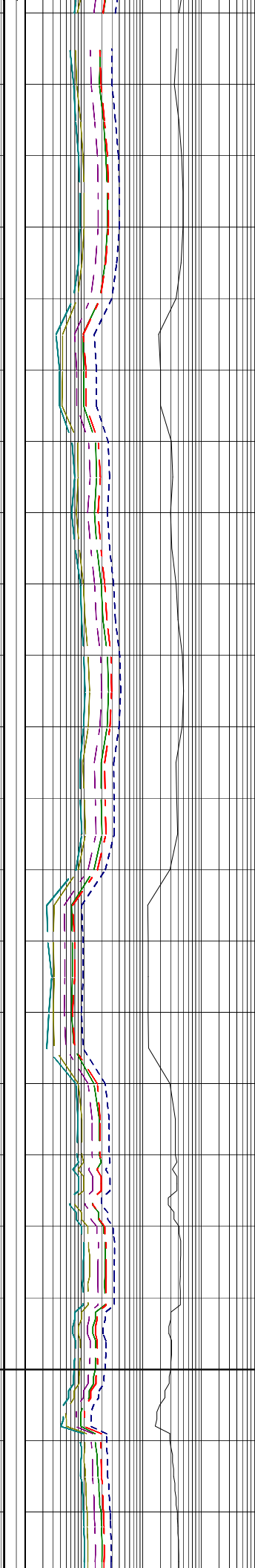
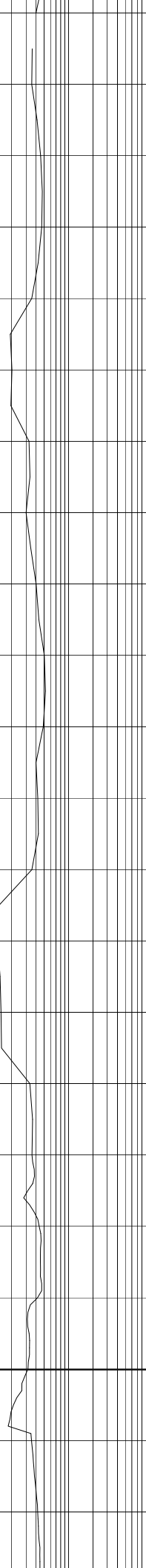
CHANGE TO 1'
DATABASE @ 3950'



80 / 4 / 6 / 7 / 3

77 / 4 / 7 / 8 / 4

79 / 4 / 6 / 7 / 4



SURVEY @ 3794.74'MDRT
INC:53.99° AZI:201.09°G

SILTSTONE:med-mod dk gry,
med brnsh gry,tr lt gry,arg,tr glauc,
tr carb spks,mod hd-occ hd,
sbbiky-sbfiss.

SURVEY @ 3825.85'MDRT
INC:55.14° AZI:201.12°G

SILTSTONE:med-mod dk gry,
med brnsh gry,tr lt gry,arg,tr glauc,
tr carb spks,tr dissem pyr,mod
hd-occ hd,sbbiky-sbfiss.

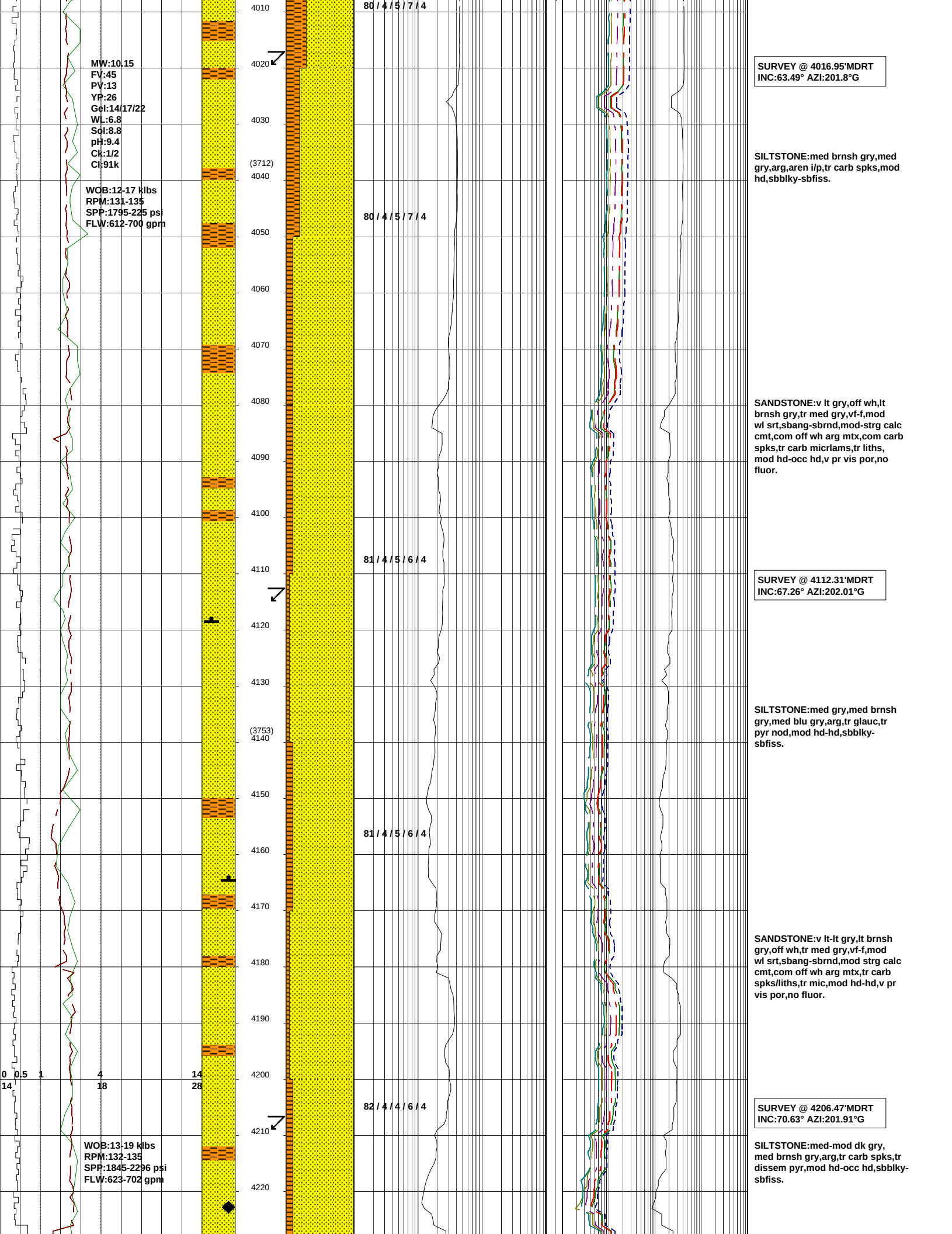
SANDSTONE:lt-med gry,off wh,vf-f,
tr med,mod wl srt,sbang-sbrnd,mod
calc cmt,off wh-med gry arg mtx,tr
glauc,tr liths,mod hd-hd,v pr vis
por,no fluor.

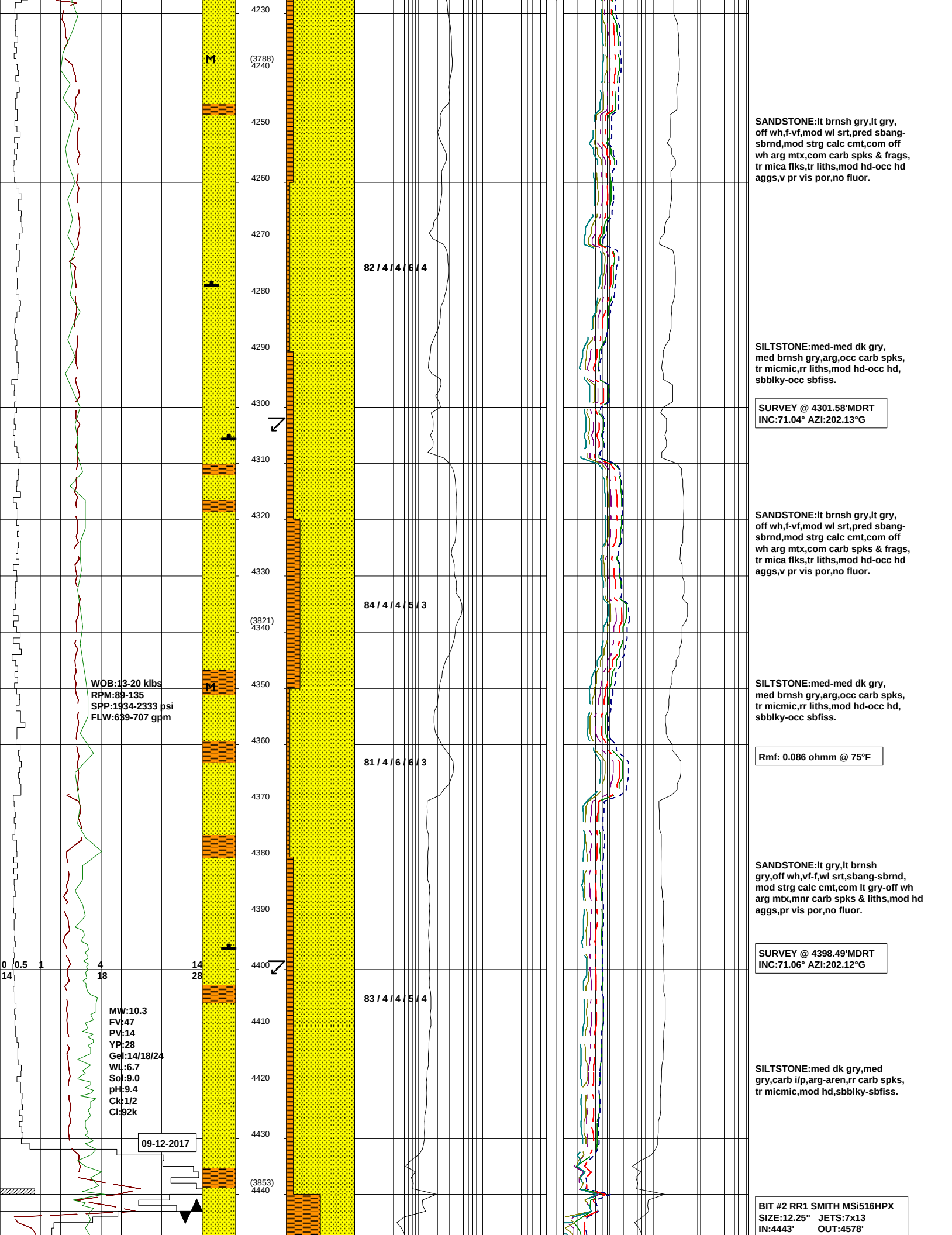
SURVEY @ 3921.22'MDRT
INC:59.24° AZI:201.22°G

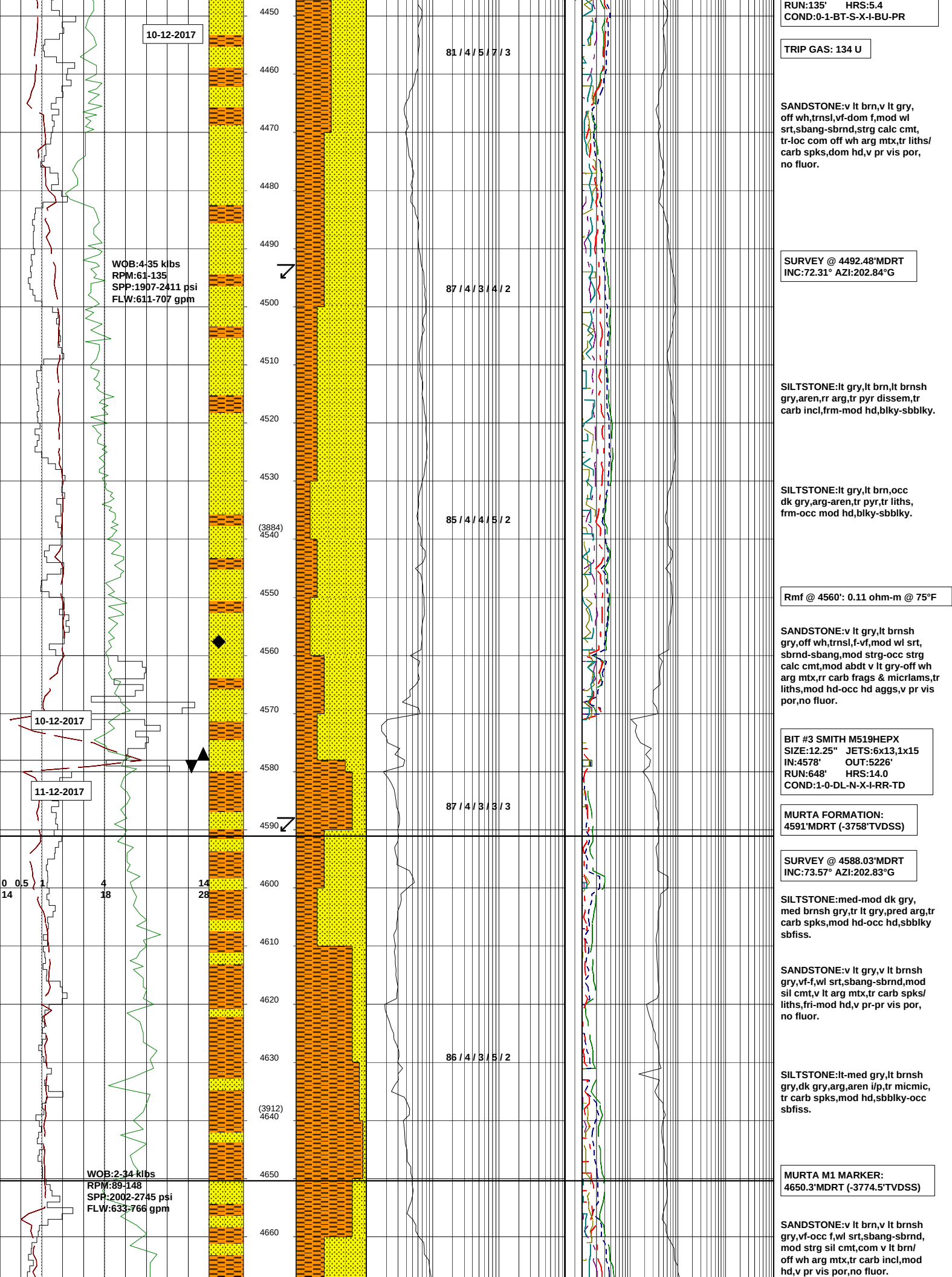
SILTSTONE:med gry,mod dk gry
i/p,brnsh gry,arg,tr aren,tr glauc,
carb spks,frm-mod hd,sbbiky-
sbfiss.

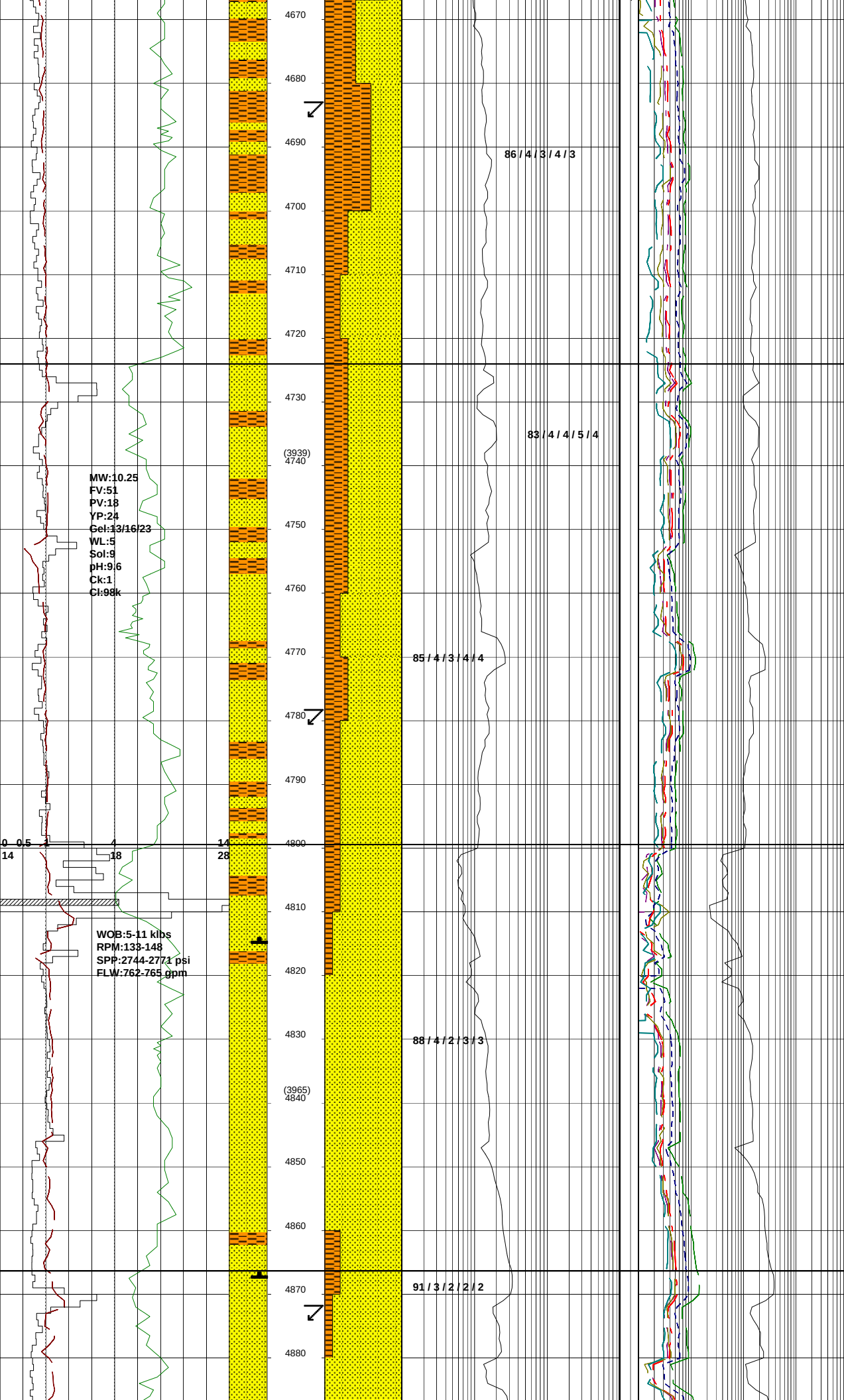
CADNA-OWIE FORMATION:
3980'MDRT (-3546'TVDSS)

SANDSTONE:v lt gry,lt brnsh gry,
off wh,vf-pred f,mod wl srt,sbang-
sbrnd,mod strg calc cmt,com off
wh arg mtx,com carb incl,tr liths,
fri-pred mod hd,v pr vis por,no
fluor.









SILTSTONE: med gry, med dk gry, arg i/p, com aren grd to intbd wl- vf SST, rr carb incl, tr liths, tr calc frags, mod hd, blkly-sbbkly, tr sbfiss.

SURVEY @ 4683.12'MDRT
INC: 74.31° AZI: 202.65°G

SANDSTONE: v lt brn-off wh, vf-f, wl srt, sbang-sbrnd, mod strg sil cmt, mntr calc cmt, com off wh arg mtb, tr calc frags, tr carb incl, mod hd, v pr-pr vis por, no fluor.

SILTSTONE: med gry, med dk gry, occ bec lt-med brnsh gry, arg i/p, com aren grd to intbd wl- vf SST, rr carb incl, tr liths, tr calc frags, mod hd, blkly-sbbkly, tr sbfiss.

MURTA M2 MARKER:
4724'MDRT (-3794.5'TVDSS)

SANDSTONE: v lt brn, lt brnsh gry, vf-f, wl srt, sbang-sbrnd, wk-mod sil cmt, com off wh arg mtb, tr liths, mntr carb spks, fri-mod hd, pr vis por, no fluor.

SANDSTONE: v lt brn, lt brnsh gry, off wh, trnsi, cl, vf-f aggs, com lse med Qtz grns, pr srt, sbang-sbrnd, wk-mod strg sil cmt, com off wh arg mtb, tr carb spks, tr pyr, tr liths, fri-mod hd, pr vis por, no fluor.

SILTSTONE: med gry-occ lt gry, lt brnsh gry, occ med dk gry, aren, mntr arg, tr micmic, mntr carb spks, occ sndy lams, mod hd, sbbkly-occ sbfiss.

SURVEY @ 4778.82'MDRT
INC: 74.58° AZI: 201.85°G

SANDSTONE: v lt brn, lt brnsh gry, off wh, cl, trnsi, vf-f, occ med lse grs, pr srt, sbang-sbrnd, wk-occ mod strg calc cmt, com off wh-lt brnsh gry arg mtb, occ carb spks, rr slty lams, tr liths, tr pyr, fri-occ mod hd aggs, pr vis por, no fluor.

MURTA M3 MARKER:
4799.4'MDRT (-3814.5'TVDSS)

SANDSTONE: v lt brn, v lt brnsh gry, off wh, vf-pred f, wl srt, sbang-sbrnd, wk-mod calc+sil cmt, com off wh arg mtb, rr carb spks, fri-mod hd, v pr-pr vis por, no fluor.

LIMESTONE: trnsi, v lt brn, micr, tr pyr vns, mod hd-hd, brit i/p, blkly.

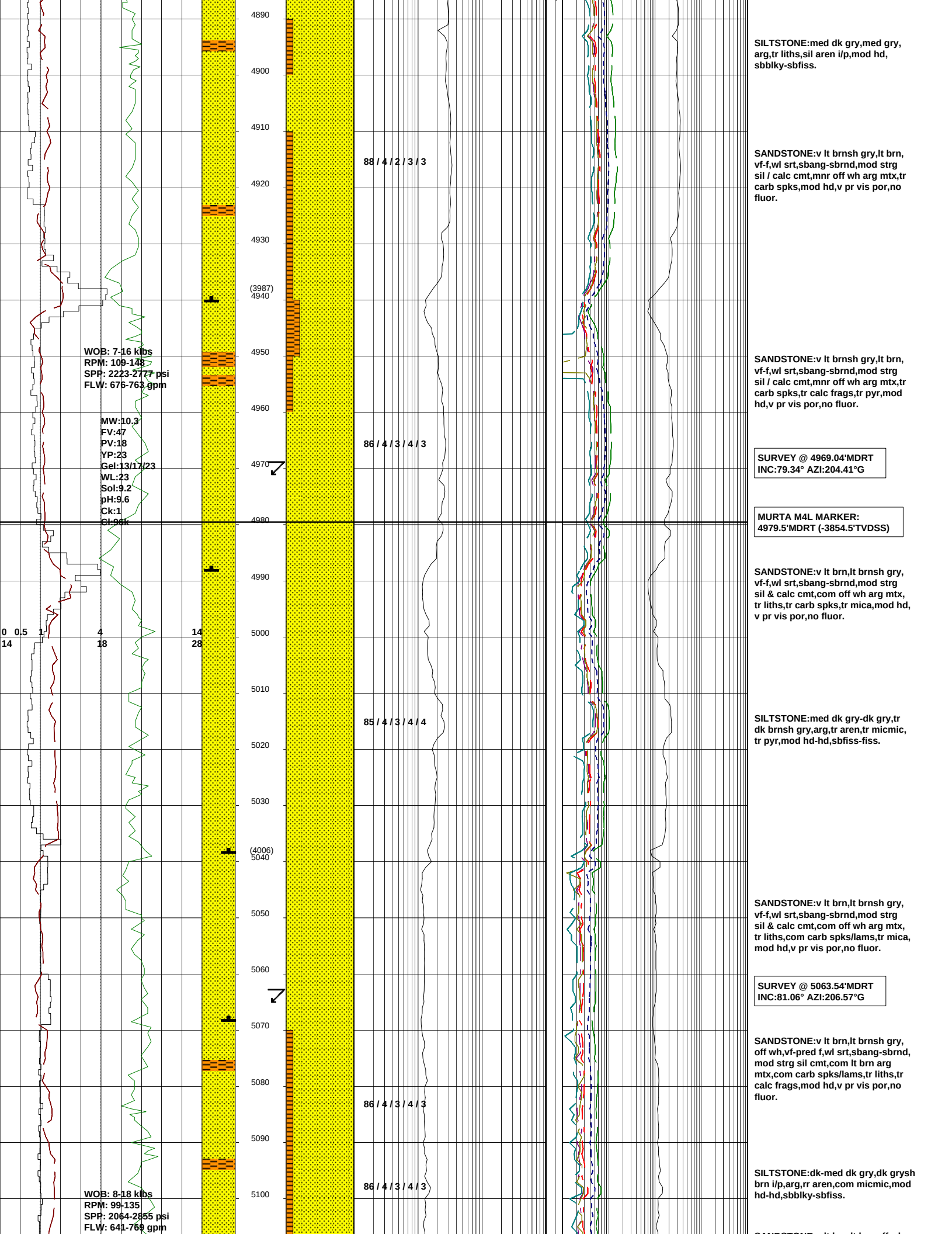
SANDSTONE: v lt brn, lt brnsh gry, trnsi i/p, vf-f, mod wl srt, sbang-sbrnd, mod strg calc / sil cmt, tr pyr, tr calc frags, tr carb spks, pred mod hd, v pr vis por, no fluor.

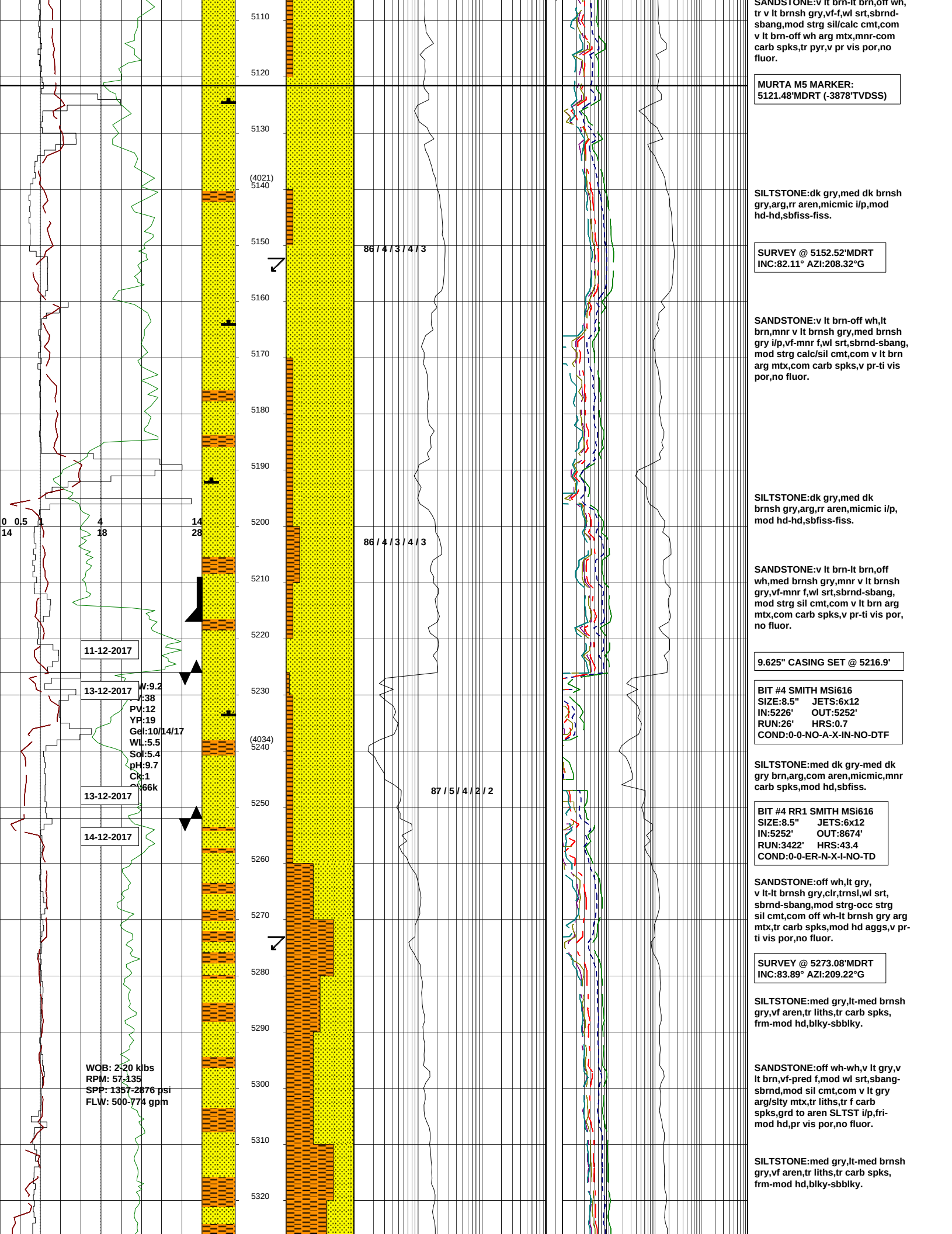
SILTSTONE: med dk gry, med gry, arg, tr liths, sil aren i/p, mod hd, sbbkly-sbfiss.

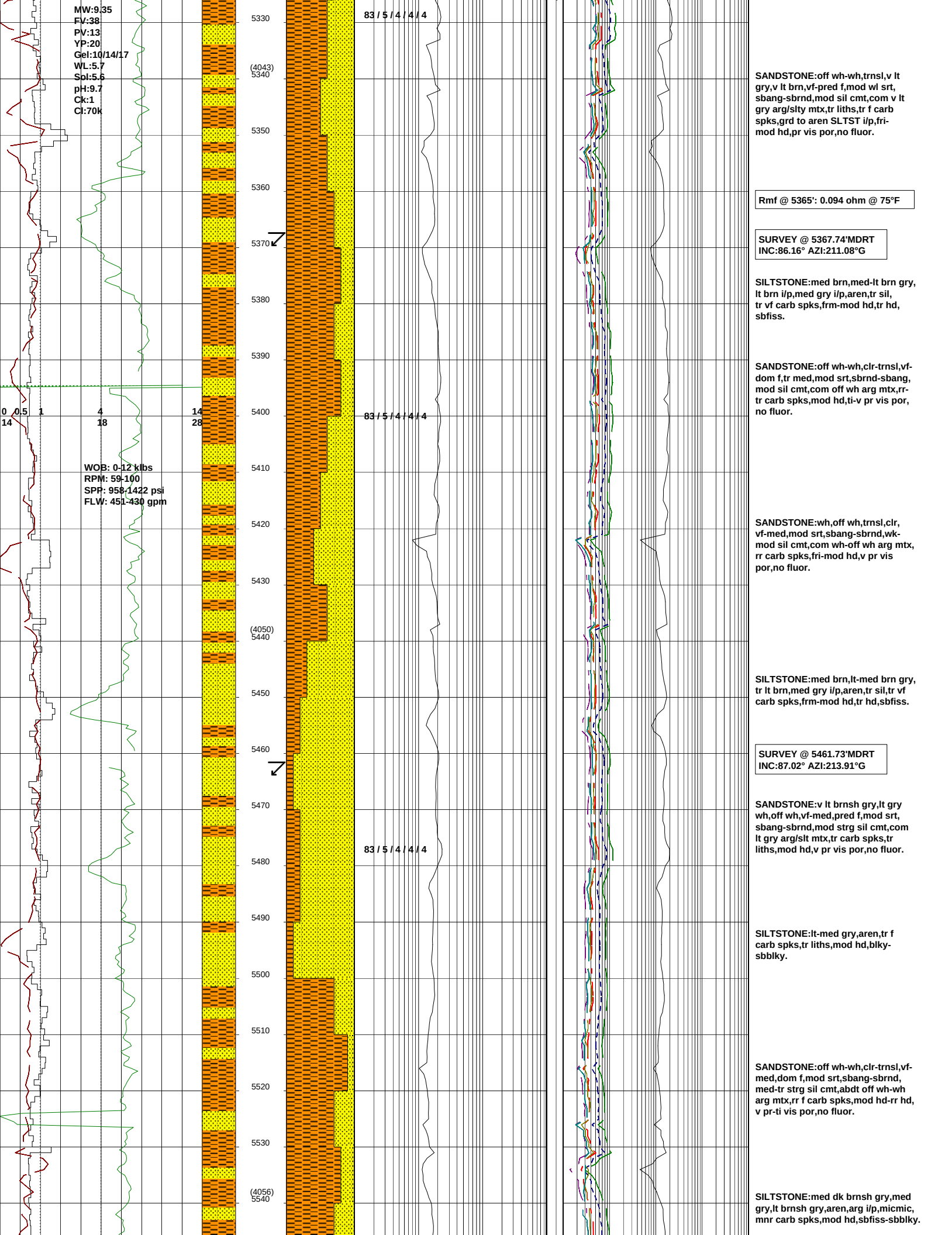
MURTA M4 MARKER:
4866.3'MDRT (-3831'TVDSS)

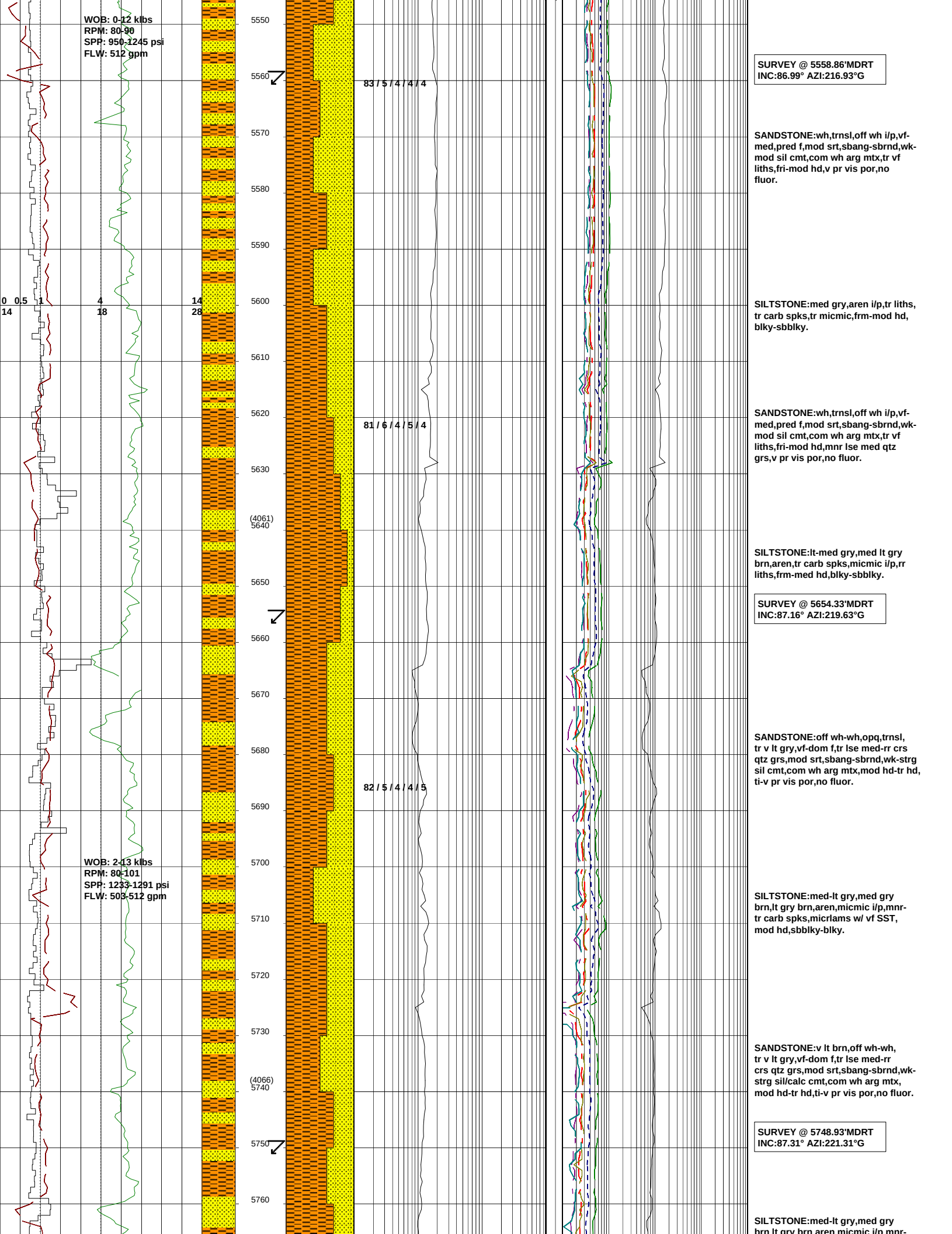
SURVEY @ 4872.84'MDRT
INC: 76.58° AZI: 202.72°G

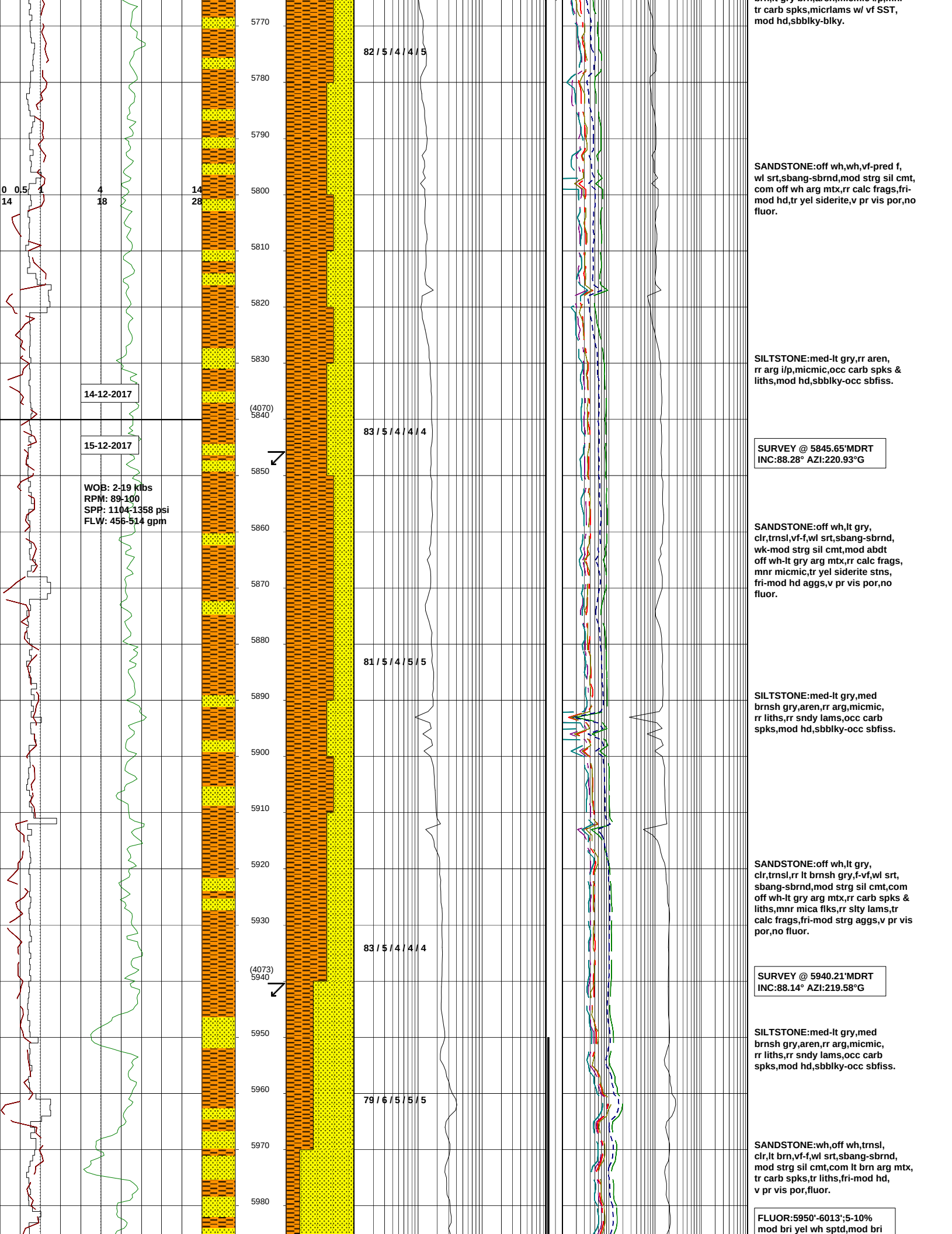
SANDSTONE: v lt brnsh gry, trnsi, vf-f, wl srt, sbang-sbrnd, mod strg sil cmt, mntr calc cmt, com lt brn arg mtb, tr lith, tr carb spks, mod hd, v pr vis por, no fluor.

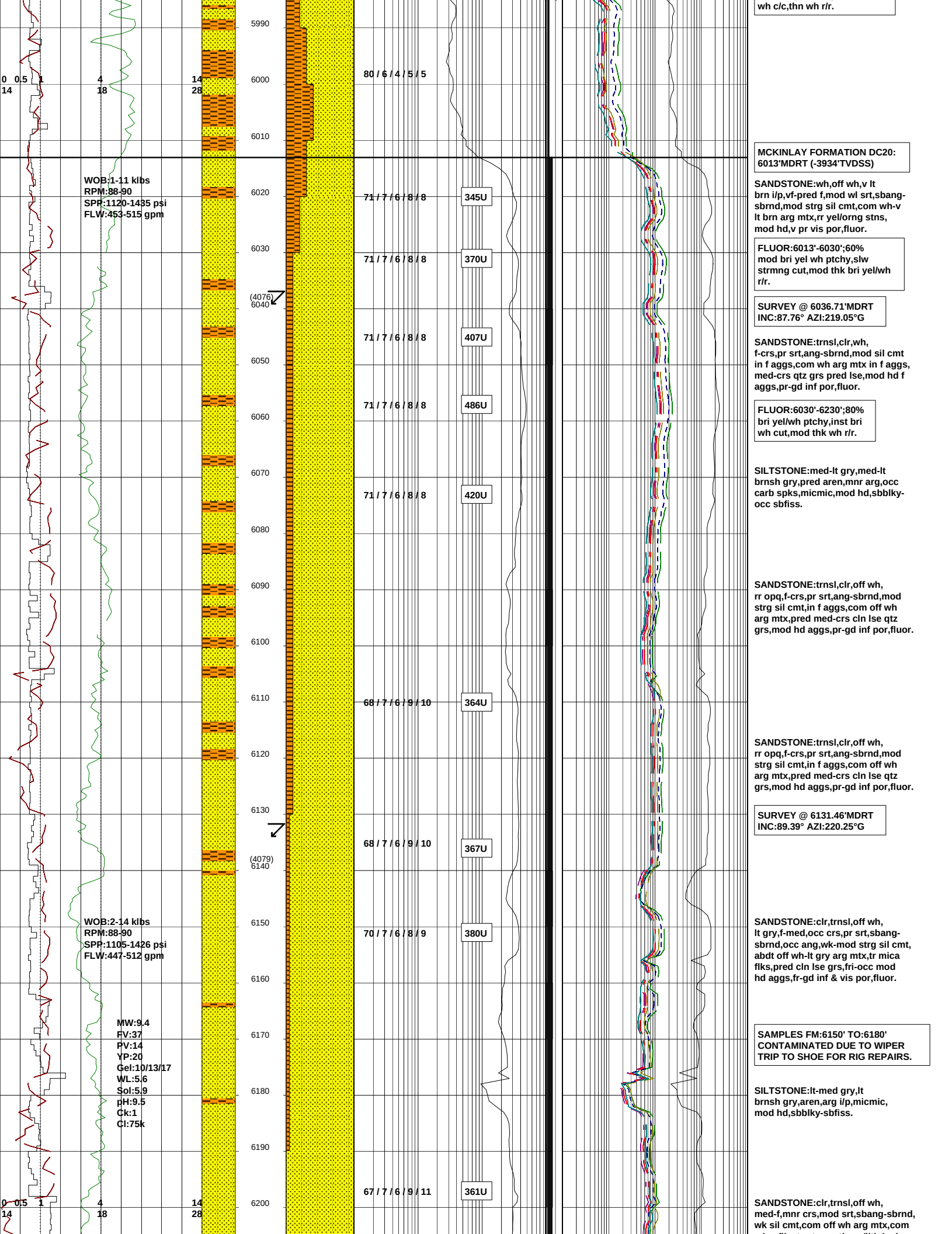


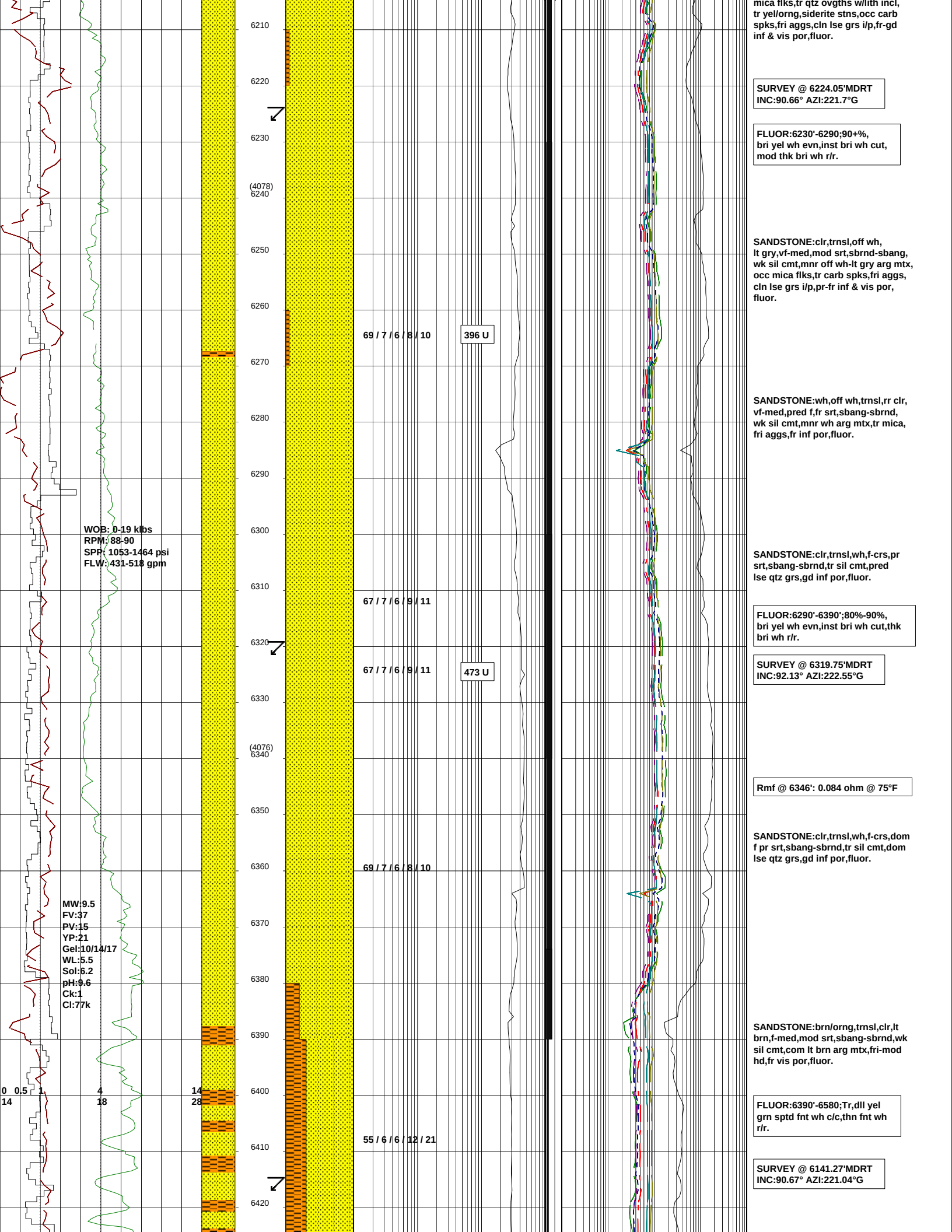


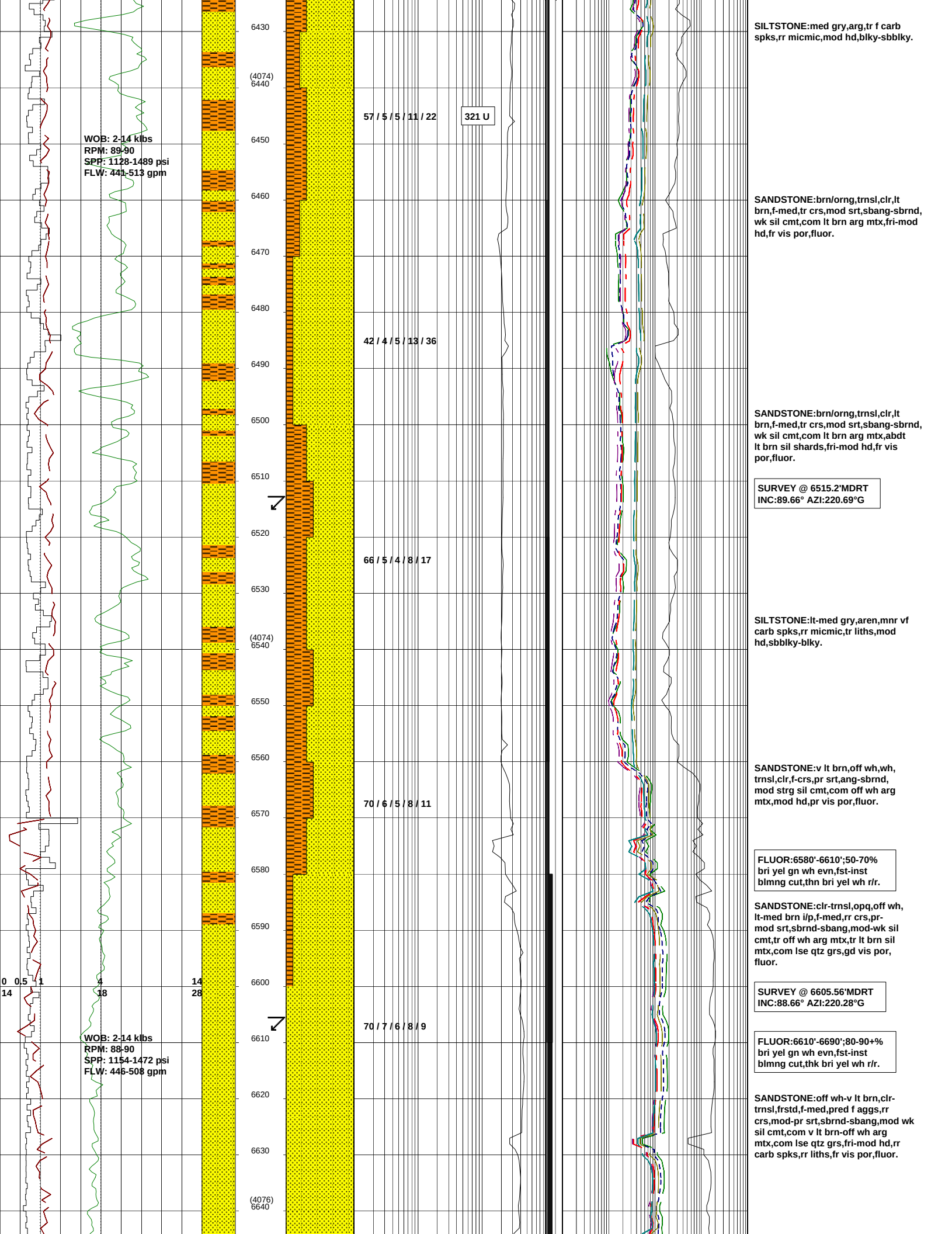


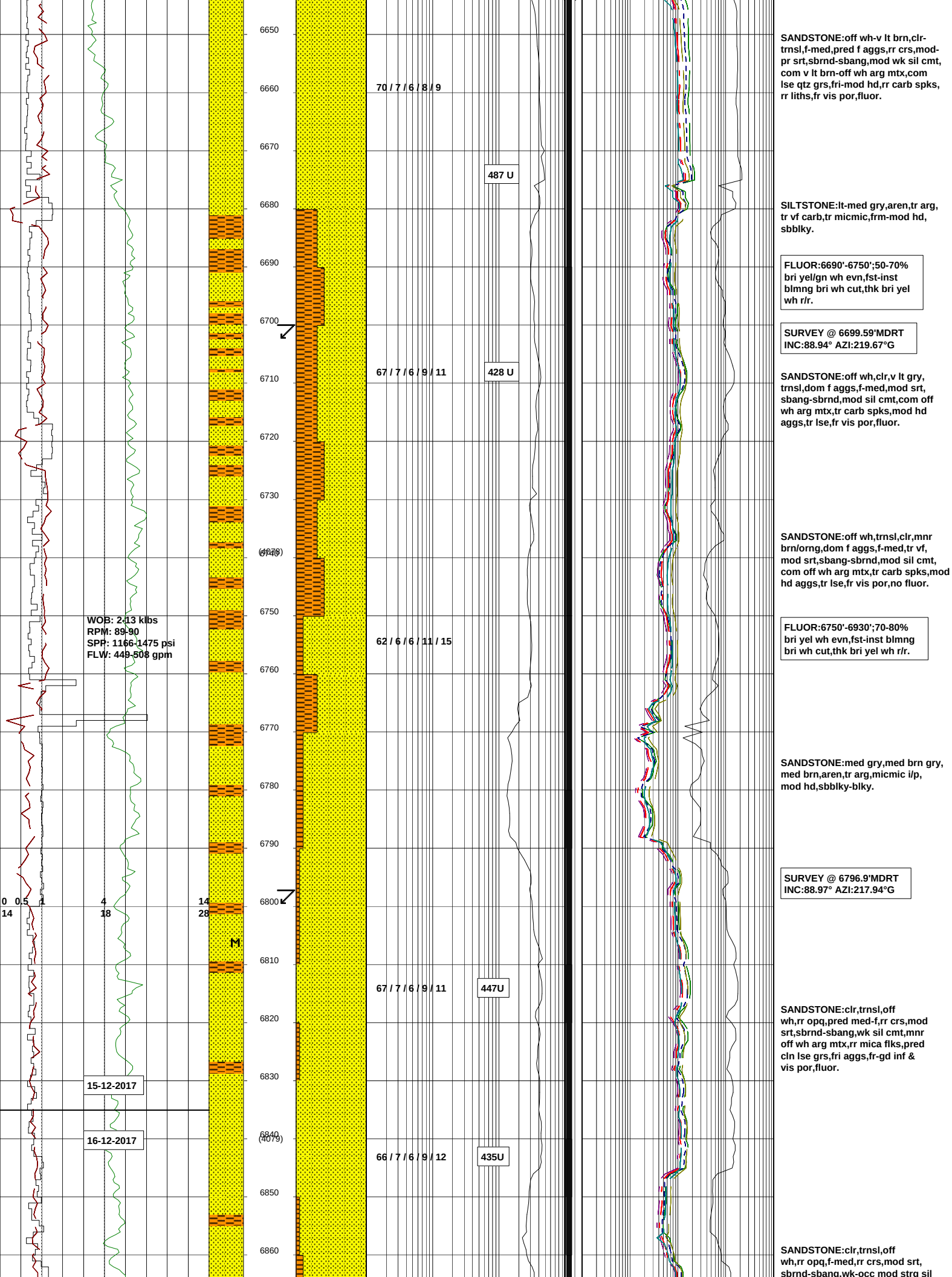












WOB: 2413 klbs
RPM: 89-90
SPP: 1166-1475 psi
FLW: 449-508 gpm

15-12-2017

16-12-2017

70 / 7 / 6 / 8 / 9

67 / 7 / 6 / 9 / 11

62 / 6 / 6 / 11 / 15

67 / 7 / 6 / 9 / 11

66 / 7 / 6 / 9 / 12

487 U

428 U

447 U

435 U

SANDSTONE: off wh-v lt brn, clr-trnsl, f-med, pred f aggs, rr crs, mod-pr srt, sbrnd-sbang, mod wk sil cmt, com v lt brn-off wh arg mtx, com lse qtz grs, fri-mod hd, rr carb spks, rr liths, fr vis por, fluor.

SILTSTONE: lt-med gry, aren, tr arg, tr vf carb, tr micmic, frm-mod hd, sbblky.

FLUOR: 6690'-6750'; 50-70% bri yel/gn wh evn, fst-inst blmng bri wh cut, thk bri yel wh r/r.

SURVEY @ 6699.59' MDRT
INC: 88.94° AZI: 219.67° G

SANDSTONE: off wh, clr, v lt gry, trnsl, dom f aggs, f-med, mod srt, sbang-sbrnd, mod sil cmt, com off wh arg mtx, tr carb spks, mod hd aggs, tr lse, fr vis por, fluor.

SANDSTONE: off wh, trnsl, clr, mnrr brn/orng, dom f aggs, f-med, tr vf, mod srt, sbang-sbrnd, mod sil cmt, com off wh arg mtx, tr carb spks, mod hd aggs, tr lse, fr vis por, no fluor.

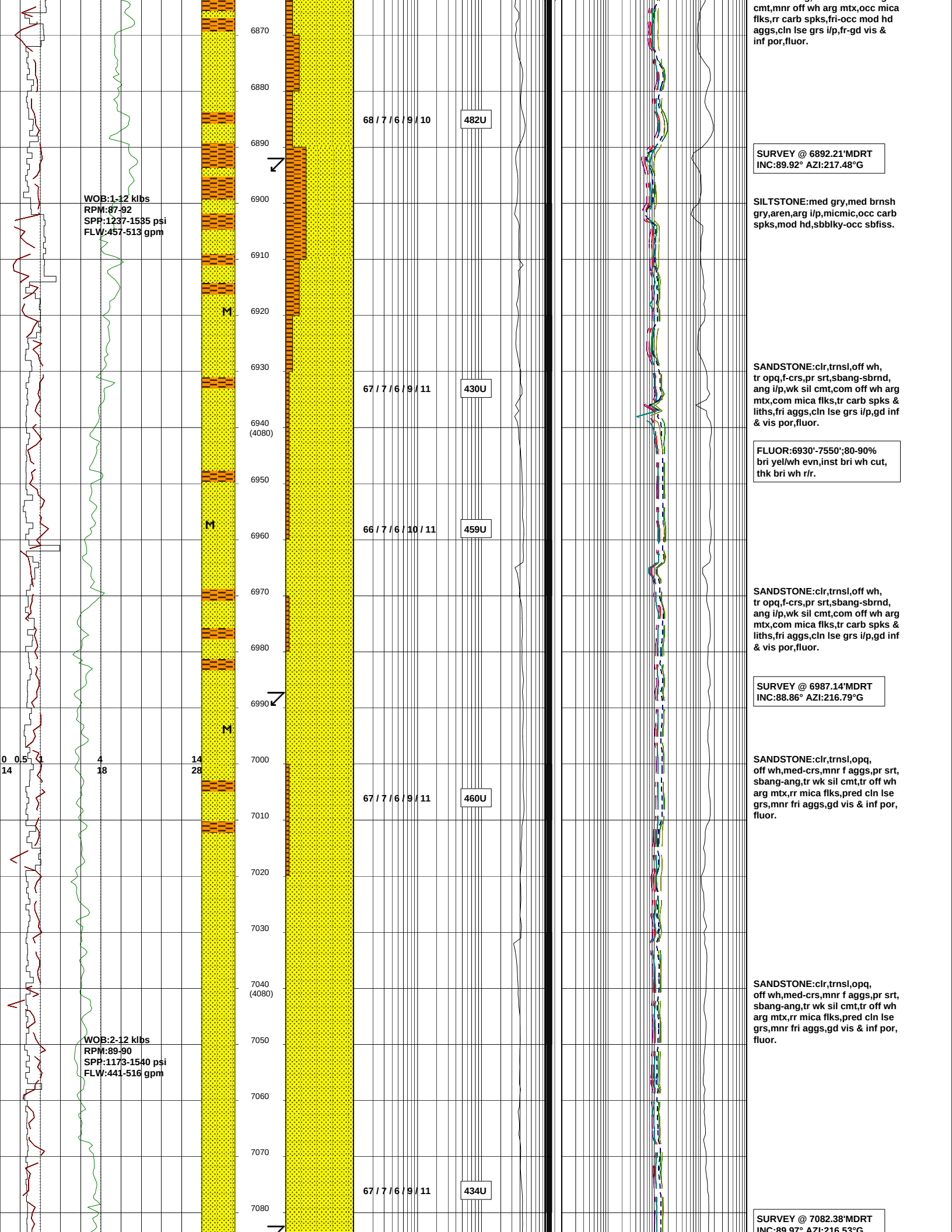
FLUOR: 6750'-6930'; 70-80% bri yel wh evn, fst-inst blmng bri wh cut, thk bri yel wh r/r.

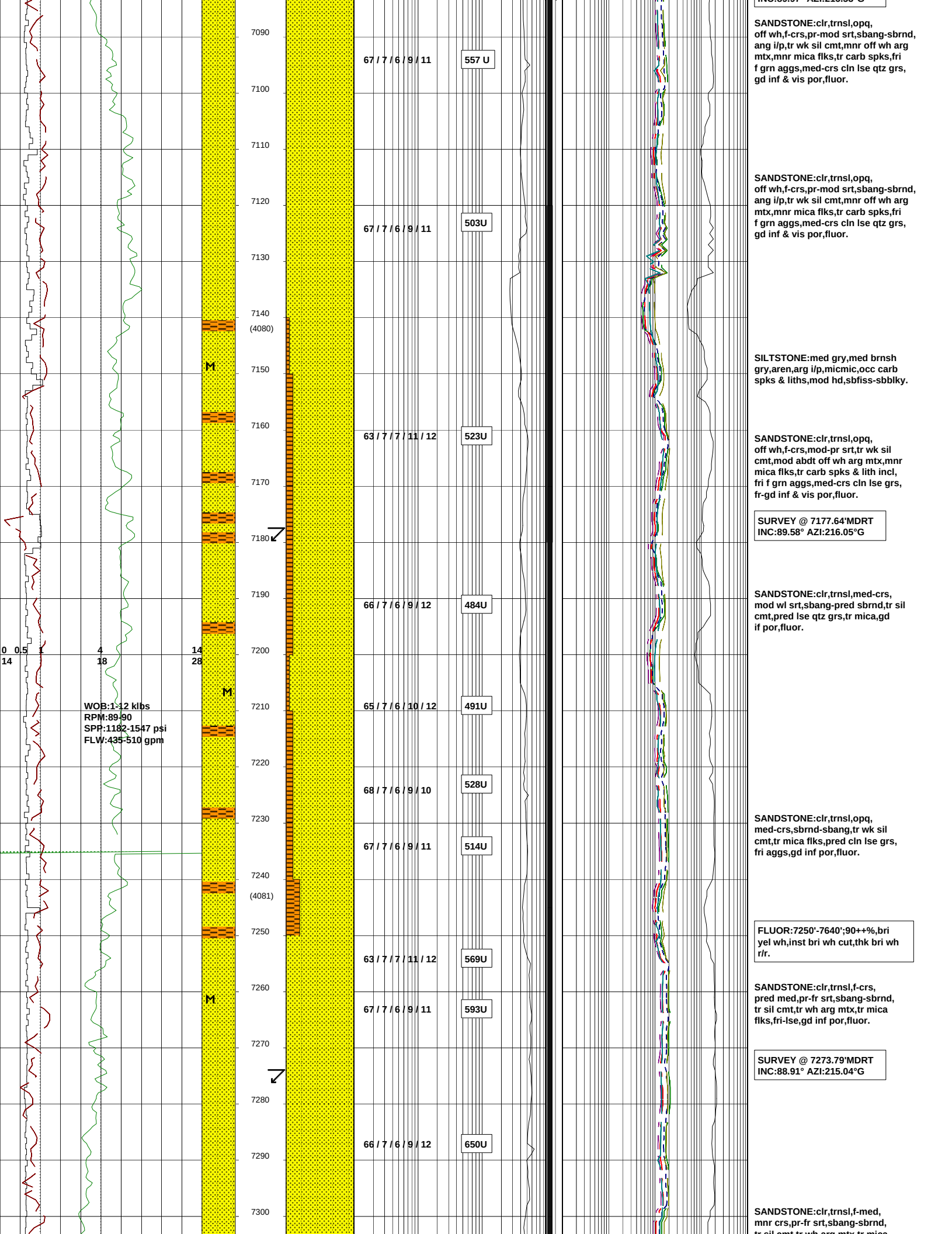
SANDSTONE: med gry, med brn gry, med brn, aren, tr arg, micmic i/p, mod hd, sbblky-blky.

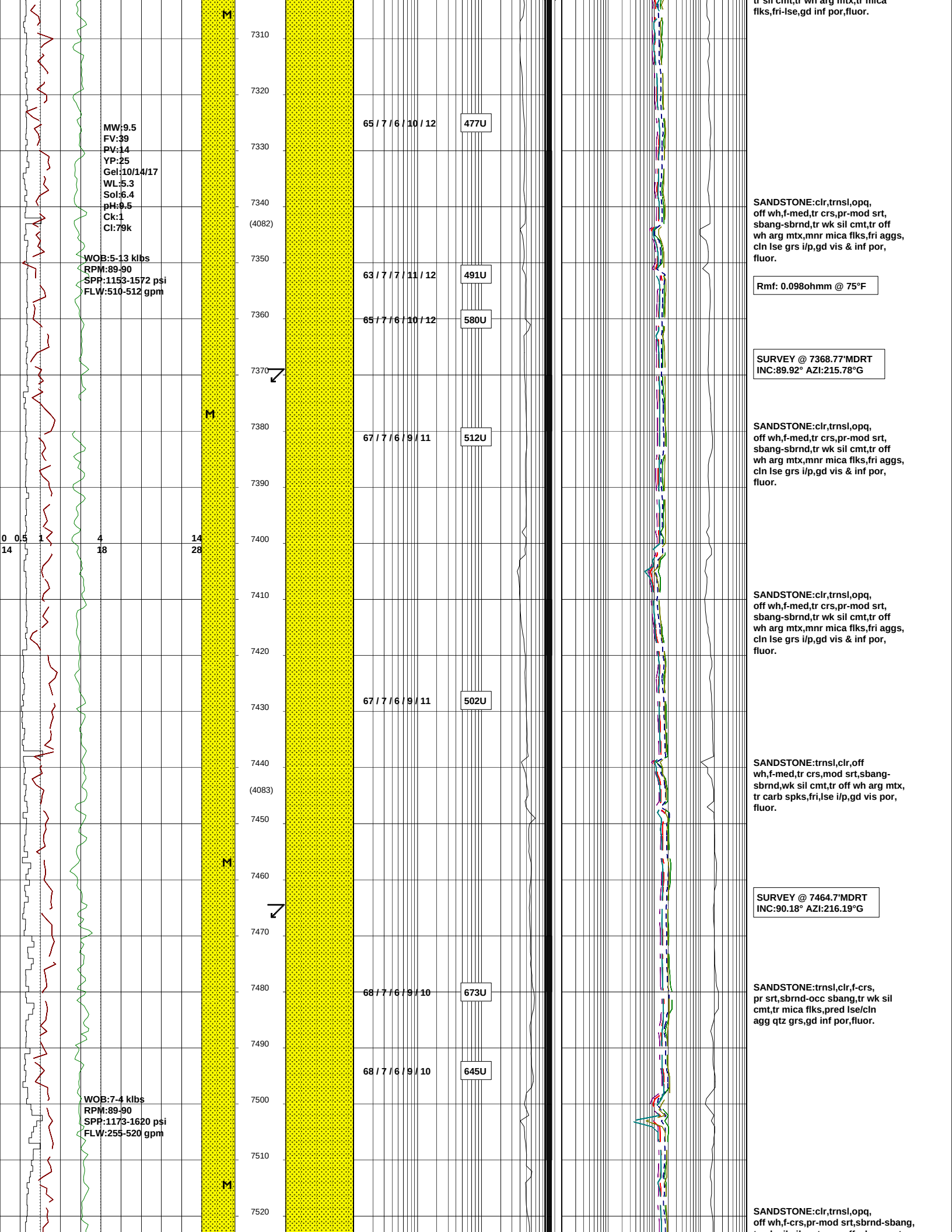
SURVEY @ 6796.9' MDRT
INC: 88.97° AZI: 217.94° G

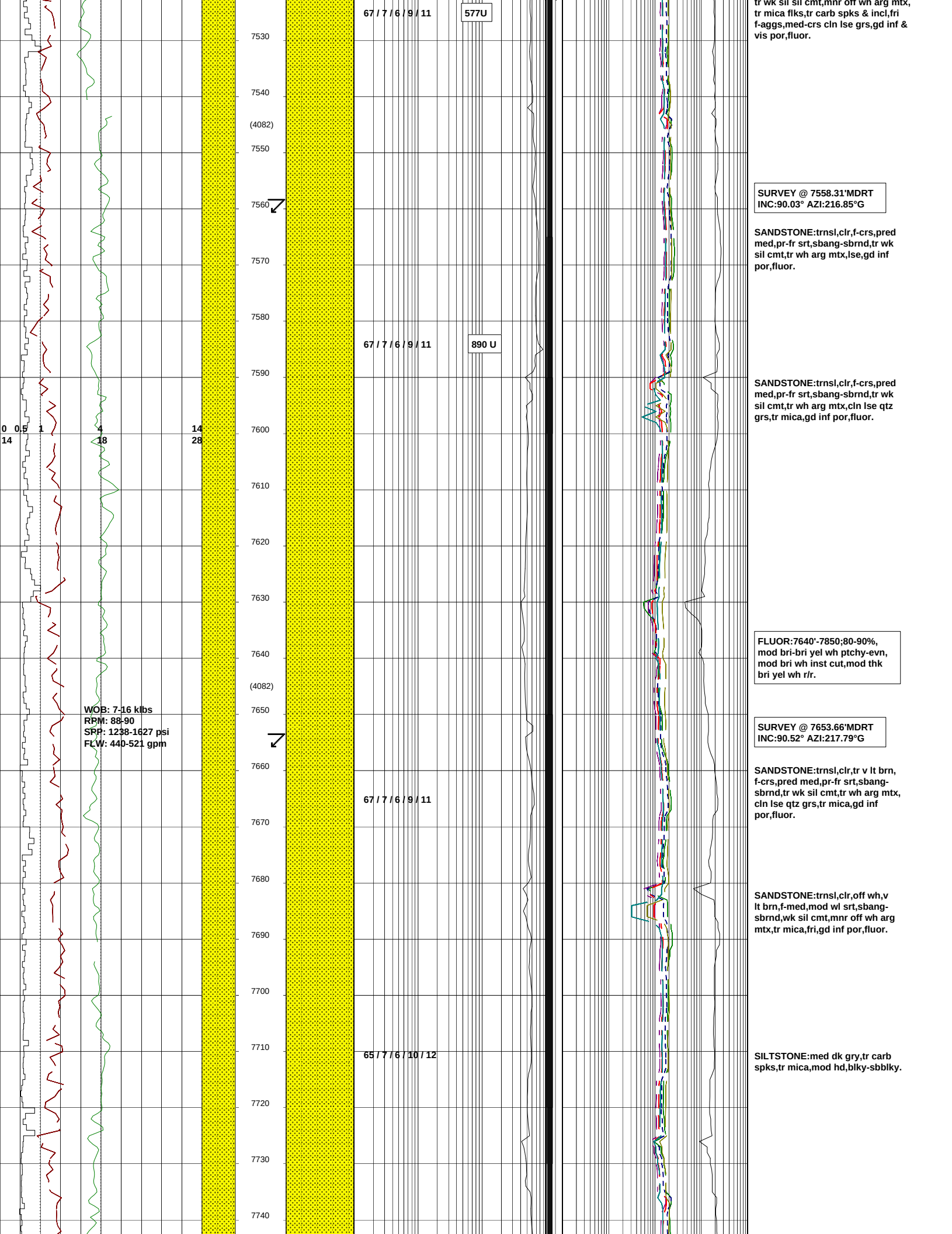
SANDSTONE: clr, trnsl, off wh, rr opq, pred med-f, rr crs, mod srt, sbrnd-sbang, wk sil cmt, mnrr off wh arg mtx, rr mica flks, pred cin lse grs, fri aggs, fr-gd inf & vis por, fluor.

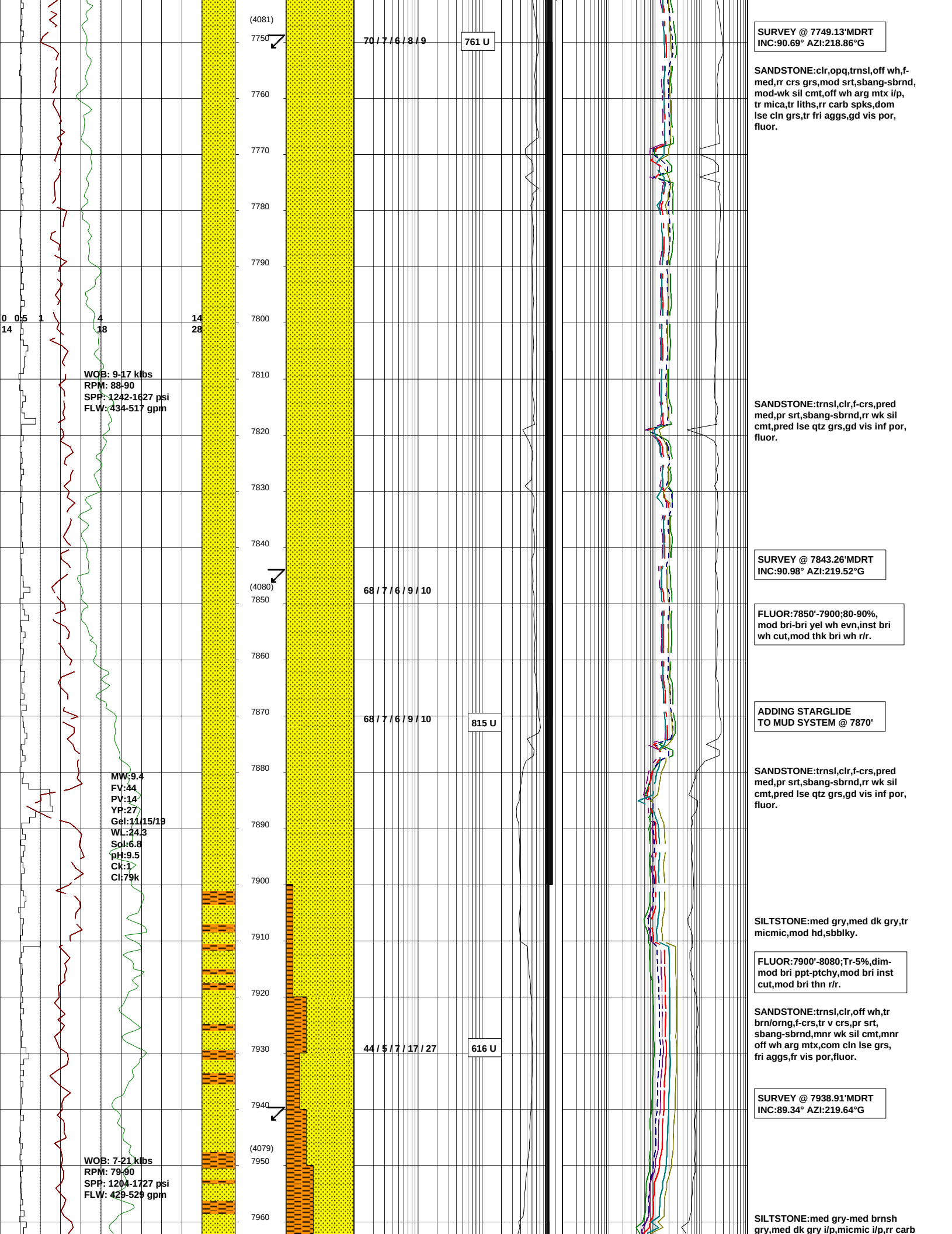
SANDSTONE: clr, trnsl, off wh, rr opq, f-med, rr crs, mod srt, sbrnd-sbang, wk-occ mod stra sil

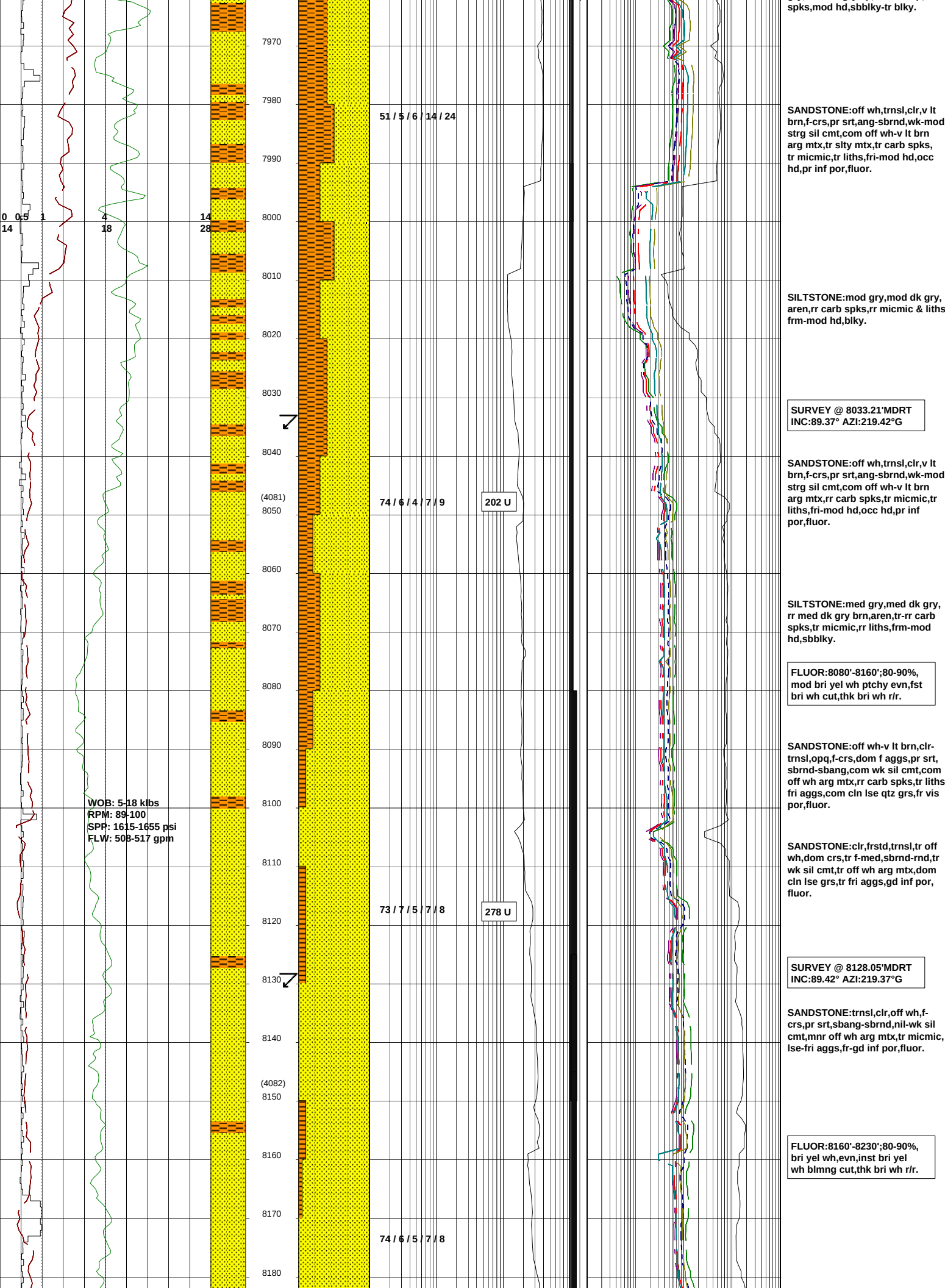












spks,mod hd,sbblky-tr blkly.

SANDSTONE:off wh,trnsl,clr,v lt brn,f-crs,pr srt,ang-sbrnd,wk-mod strg sil cmt,com off wh-v lt brn arg mtz,tr slty mtz,fr carb spks, tr micmic,tr liiths,fr-mod hd,occ hd,pr inf por,fluor.

SILTSTONE:mod gry,mod dk gry, aren,rr carb spks,rr micmic & liiths, frm-mod hd,blkly.

SURVEY @ 8033.21'MDRT
INC:89.37° AZI:219.42°G

SANDSTONE:off wh,trnsl,clr,v lt brn,f-crs,pr srt,ang-sbrnd,wk-mod strg sil cmt,com off wh-v lt brn arg mtz,rr carb spks,fr micmic,tr liiths,fr-mod hd,occ hd,pr inf por,fluor.

SILTSTONE:med gry,med dk gry, rr med dk gry brn,aren,rr-rr carb spks,fr micmic,rr liiths,frm-mod hd,sbblky.

FLUOR:8080'-8160';80-90%, mod bri yel wh ptchy evn,fst bri wh cut,thk bri wh r/r.

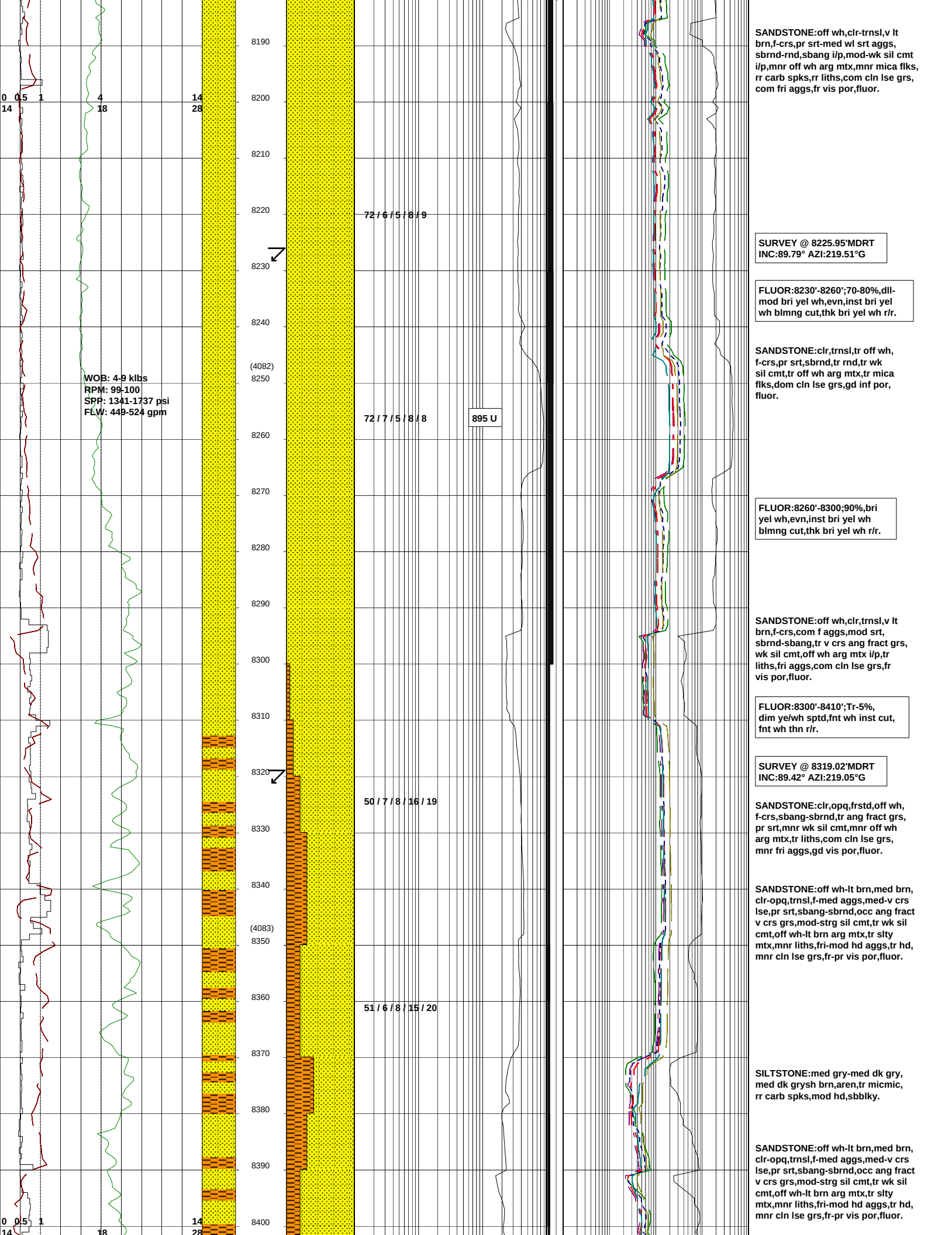
SANDSTONE:off wh-v lt brn,clr-trnsl,opq,f-crs,dom f aggs,pr srt, sbrnd-sbang,com wk sil cmt,com off wh arg mtz,rr carb spks,fr liiths, fri aggs,com cln lse qtz grs,fr vis por,fluor.

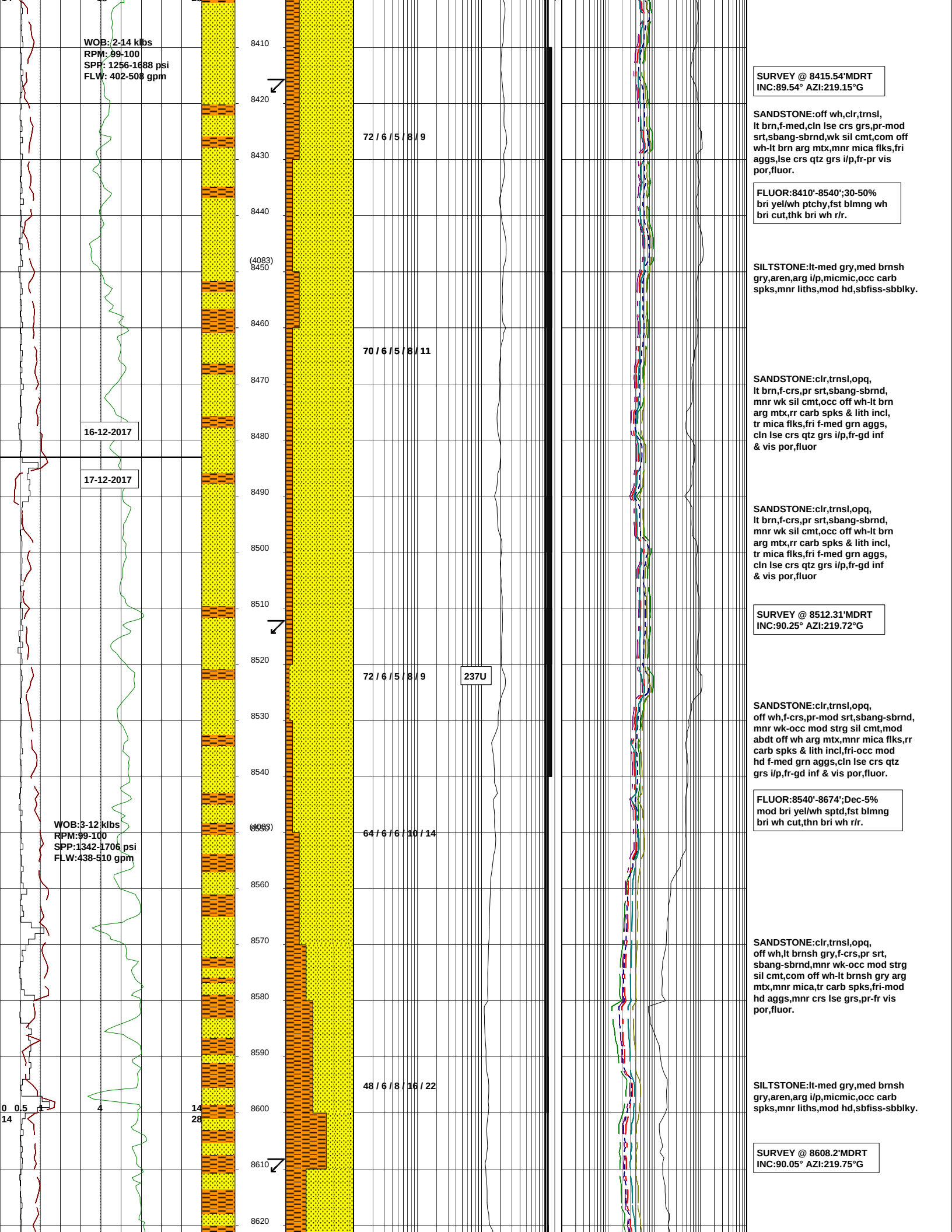
SANDSTONE:clr,frstd,trnsl,fr off wh,dom crs,fr f-med,sbrnd-rnd,fr wk sil cmt,fr off wh arg mtz,dom cln lse grs,fr fri aggs,gd inf por, fluor.

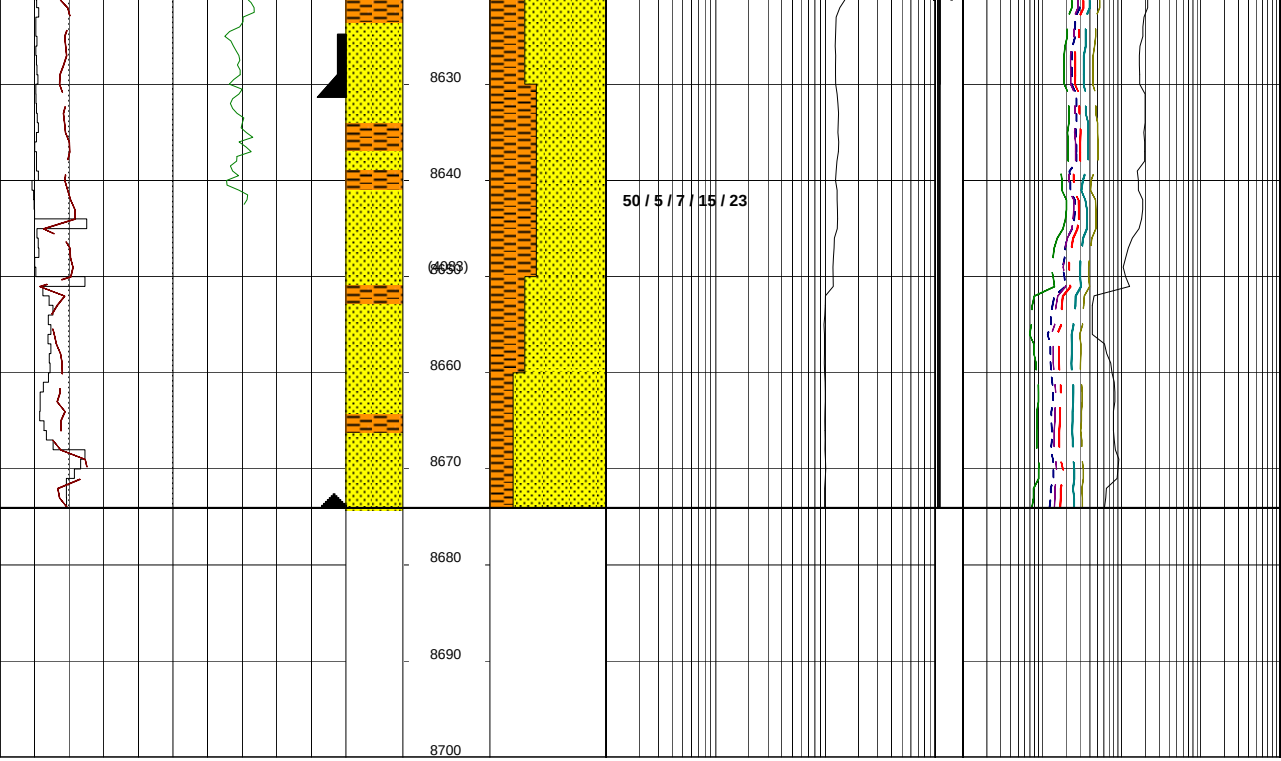
SURVEY @ 8128.05'MDRT
INC:89.42° AZI:219.37°G

SANDSTONE:trnsl,clr,off wh,f-crs,pr srt,sbang-sbrnd,nil-wk sil cmt,mnr off wh arg mtz,fr micmic, lse-fri aggs,fr-gd inf por,fluor.

FLUOR:8160'-8230';80-90%, bri yel wh,evn,inst bri yel wh blmng cut,thk bri wh r/r.







5.5" LINER
SET @ 8631.35'MDRT

SANDSTONE:clr,trnsl,opq,
off wh,lt brnsh gry,f-crs,pr srt,
sbang-sbrnd,mnr wk-occ mod strg
sil cmt,com off wh-lt brnsh gry arg
mtx,mnr mica,tr carb spks,fri-mod
hd aggs,mnr crs lse grs,pr-fr vis
por,fluor.

WIPER TRIP FROM 8674' TO 7754'

MCKINLAY 14 REACHED TD
AT 02:45hrs ON 17-12-2017